

Anterior knee pain

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The New York Knicks announced today that Carmelo Anthony will have season-ending left knee surgery. The procedure, which will be performed by Team Orthopedist Dr. Answorth Allen, includes a left knee patella tendon debridement and repair.

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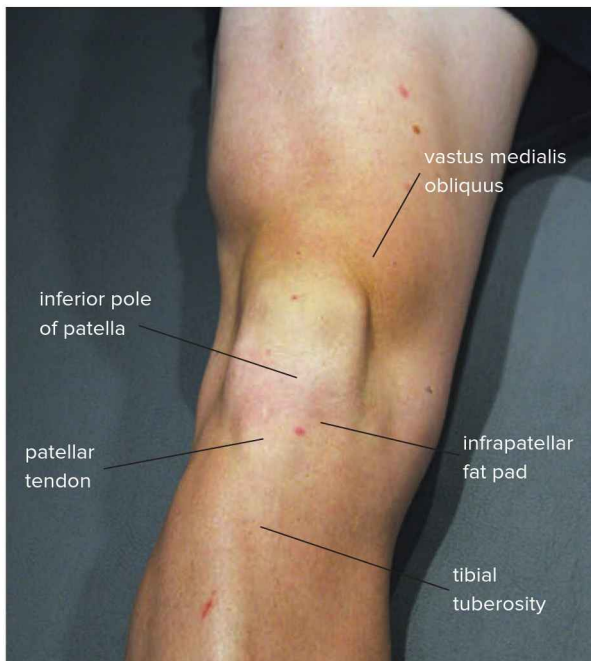
Anterior knee pain is the most common presenting symptom in many physiotherapy and sports physician practices.^{1, 2} It contributes substantially to the 20–40% of family practice consultations that relate to the musculoskeletal system.³ Anterior knee anatomy is shown in Figure 36.1.

In this chapter, we:

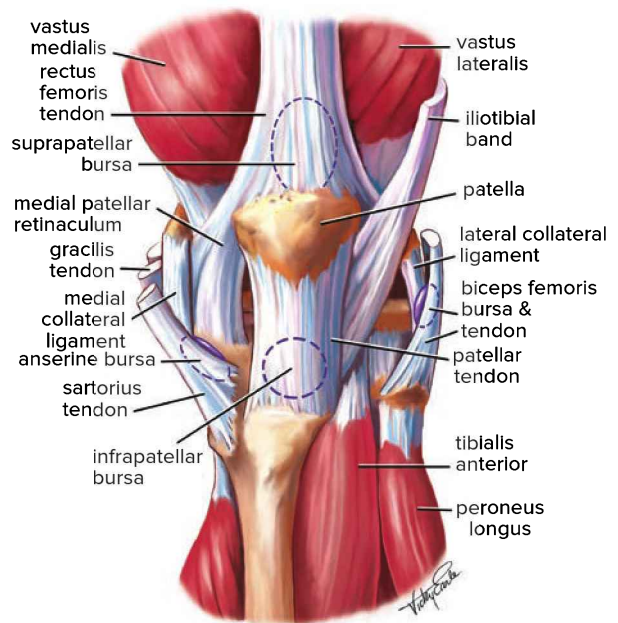
- outline the clinical approach to assessing the patient with anterior knee pain, particularly with a view to

distinguishing the common conditions—patellofemoral pain (PFP) and patellar tendinopathy

- detail contemporary management integrating high level evidence (Chapter 2) with the best of clinical experience
- discuss less common causes of anterior knee pain such as fat pad impingement, which may mimic features of both PFP and patellar tendinopathy
- highlight new research investigating PFP in adolescents, patellar tendinopathy and patellofemoral osteoarthritis (OA).



(a)



(b)

Figure 36.1 Anterior knee (a) Surface anatomy (b) Anatomy

CLINICAL APPROACH

Distinguishing between PFP and patellar tendinopathy as a cause of anterior knee pain can be difficult. Rarely, both conditions may be present. Causes of anterior knee pain are listed in Table 36.1.

History

A number of important factors can be elicited from the history of a athlete presenting with anterior knee pain. These include the specific location of the pain, the nature of aggravating activities, the history of the onset and behaviour of the pain, as well as any associated clicking, giving way or swelling.

Although it may be difficult for the patient with anterior knee pain to be specific about the location of pain, this symptom provides an important clue as to the likely diagnosis. For example, retropatellar or peripatellar pain suggests that the patellofemoral joint (PFJ) is a likely culprit, lateral pain localised to the lateral femoral epicondyle increases the likelihood of iliotibial band syndrome as the diagnosis (Chapter 37). Patellar tendon pain will be localised to the inferior pole of the patella and the patient can point to it with a single finger. The pain remains localised both at rest and on loading with the decline squat. The location of pain (and source) appear to be associated with different changes to motor drive, indicating that the rehabilitation approaches are likely to be different.⁴

The onset of PFP is typically insidious and unrelated to an acute injury. However, it may present secondary to an acute traumatic episode (e.g. falling on the knee) or consequent to another knee injury (e.g. meniscal, anterior cruciate ligament) or knee surgery. The patient presents with a diffuse ache, which is usually exacerbated by

PFJ-loaded activities, such as ascending/descending stairs or running. Sometimes PFP is aggravated by prolonged sitting ('movie-goer's knee'), but sitting also tends to aggravate patellar tendinopathy symptoms and therefore is not diagnostic of PFP. Pain during running that gradually worsens is more likely to be of patellofemoral origin, whereas pain that occurs at the start of activity, settles after warm-up and returns after activity is more likely to be patellar tendinopathy. While the pathologies of patellar tendinopathy and PFP may co-exist, in practice they rarely present as co-existing clinical entities. Table 36.2 is an aid to differentiate PFP and patellar tendinopathy based on clinical history and presentation.

The type of activity that aggravates the anterior knee pain also aids diagnosis. Consider two contrasting scenarios that a patient may report as causing pain at the infrapatellar region. In one case, precipitating activities, such as basketball, volleyball, high, long or triple jumps, that involve repetitive loading (specifically energy storage and release) of the patellar tendon suggest the diagnosis of patellar tendinopathy. On the other hand, if a freestyle swimmer reported pain following tumble turning or vigorous kicking in the pool, where there had been no energy storage and release load on the tendon but a forceful extension of the knee, the practitioner should suspect an irritated fat pad. The mechanism of injury and the aggravating features are critical to an accurate diagnosis.

A history of crepitus may suggest PFP. A feeling that the patella moves laterally at certain times suggests patellofemoral instability. An imminent feeling of giving way may be associated with patellar subluxation, PFP or meniscal abnormality. Although frank, dramatic giving way is usually associated with anterior cruciate ligament

Table 36.1 Causes of anterior knee pain

Common	Less common	Not to be missed
Patellofemoral pain Patellar tendinopathy	Synovial plica Pre-patellar bursitis Quadriceps tendinopathy Infrapatellar bursitis Patellofemoral instability Fat pad impingement Sinding-Larsen–Johansson lesion in adolescents (Chapter 44) Tenoperiostitis of upper tibia Stress fracture of the patella Osgood–Schlatter lesion in adolescents (Chapter 44)	Referred pain from the hip Osteochondritis dessicans Slipped capital femoral epiphysis Perthes' disease Tumour (especially in the young)

Table 36.2 Comparison of the clinical features of patellofemoral pain and patellar tendinopathy

Signs	Patellofemoral pain	Patellar tendinopathy
Onset	Running, steps/stairs, hills, any weight-bearing activities involving knee flexion	Activities involving jumping and/or changing direction
Pain	Non-specific or vague; may be medial, lateral or infrapatellar; aggravated by activities that load the PFJ	At the inferior pole of patella, aggravated by energy storage loads such as jumping
Tenderness	Usually medial or lateral facets of patella but may be tender in infrapatellar region; may have no pain on palpation due to areas of patella being inaccessible	Localised to inferior pole of patella; this tenderness can exist in jumping athletes without patellar tendinopathy
Swelling	May have small effusion, swelling, suprapatellar or infrapatellar	Tendon may be increased in thickness; no joint swelling
Clicks/clunks	Occasional	No
Crepitus	Occasionally under patella	No
Giving way	Rarely, due to quadriceps weakness/inhibition or subluxation	Rarely, due to quadriceps weakness/inhibition
Knee range of motion	May be decreased in severe cases but usually normal	Normal
Quadriceps contraction in extension	Can be painful, often normal	Often painful
PFJ movement	May be restricted in any direction. Commonly restricted medial glide due to tight lateral structures	Normal PFJ biomechanics
Functional testing	Squats, stairs may aggravate. PFJ taping should decrease pain	Decline squats aggravate pain; PFJ taping has less effect

(ACL) instability (Chapter 35), giving way resulting from pain-induced muscle inhibition is not uncommon in people with various types of anterior knee pain.

A history of previous knee injury or surgery may be important; PFP is a well-recognised complication of posterior cruciate ligament injury and following anterior cruciate ligament reconstruction (Chapter 35). Patellar tendinopathy may be present after ACL reconstruction using bone-patellar tendon-bone grafts.

Knee pain and/or effusion may result in inhibition of the quadriceps (reduced magnitude and delayed timing of electromyography (EMG) onset) commonly termed 'pain-induced muscle inhibition'. This inhibition appears to be more profound in the vastus medialis obliquus (VMO), especially at smaller knee effusion volumes.^{5, 6} Preferential inhibition of the VMO has potential to set up an imbalance in the medial and lateral forces on the patella, predisposing to PFP.⁷ Significant knee swelling is rare in atraumatic anterior knee pain and generally suggests additional intra-articular abnormality. However, a small effusion may be present with PFP.

Previous treatment and the patient's response to it should be noted. If treatment was unsuccessful, it is essential to determine whether the failure was due to

incorrect diagnosis, inappropriate treatment or poor patient adherence.

Physical examination

The primary aim of the clinical assessment is to determine the most likely cause of the patient's pain. It is critical to reproduce the patient's anterior knee pain as location of tenderness and aggravating factors are key to the differential diagnosis. This is usually done with either a double or single leg squat (Fig. 36.2c). A squat done on a decline board (Fig. 36.2d).⁸ is very painful in people with patellar tendinopathy. It is essential to palpate the anterior knee carefully to determine the site of maximal tenderness, although palpation soreness by itself is not diagnostic; for example, the patellar tendon can be tender on palpation in asymptomatic athletes.⁹

Many clinical tests outlined below have not been evaluated for their clinical utility. Those that have been evaluated for PFP and patellar tendinopathy are outlined in Tables 36.3 and 36.4. The best tests to identify people with PFP are pain during activities (squatting, stairs, prolonged sitting) and pain during an isometric quadriceps contraction, although these tests fall well below the recommended cut offs for a diagnostic test (LR + >10).¹⁰ A combination of these tests may provide greater diagnostic accuracy.¹⁰

Examination includes the following.

1. Observation
 - a. standing (Fig. 36.2a)
 - b. walking
 - c. supine (Fig. 36.2b)
2. Functional tests
 - a. double then single-leg squat (Fig. 36.2c)
 - b. step-up/step-down
 - c. jump/hop
 - d. lunge
 - e. single-leg decline squat (Fig. 36.2d)



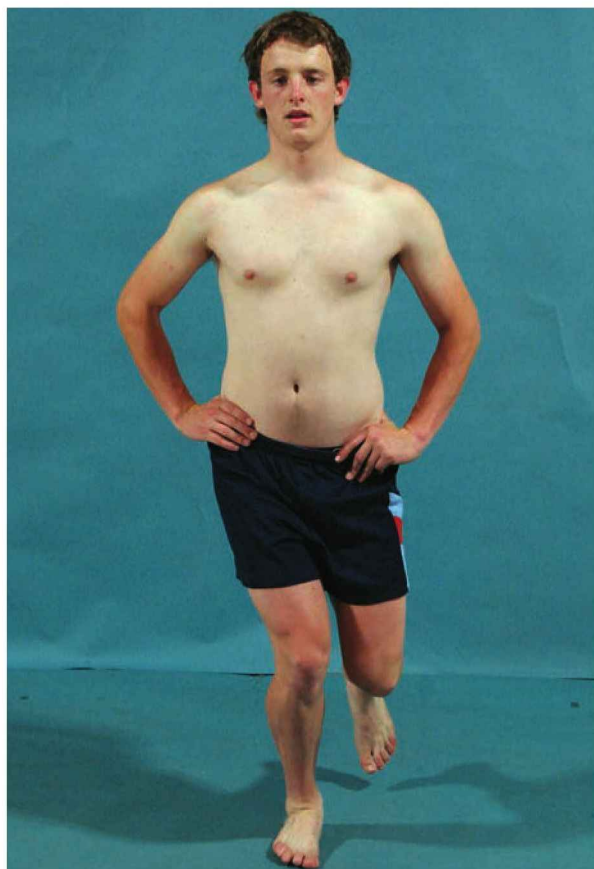
(a)

Figure 36.2 Examination of the anterior knee

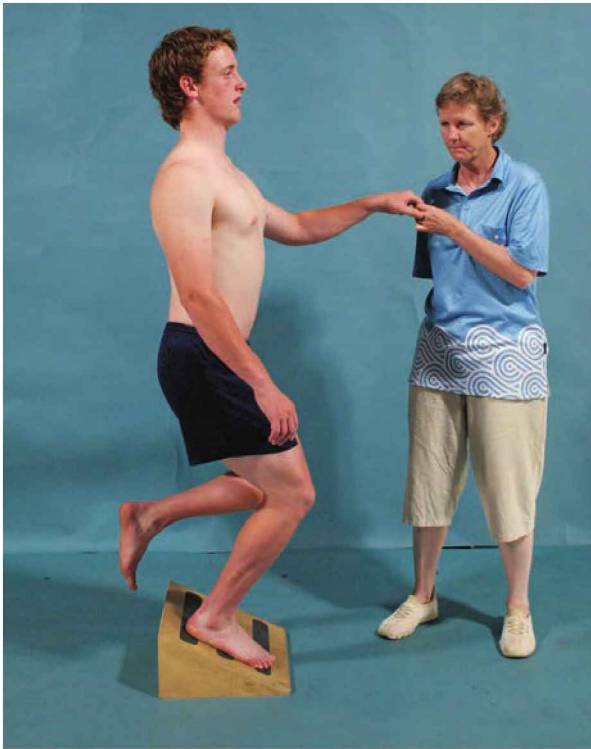
(a) Observation—standing. Observe the patient from the front to examine lower limb alignment including femoral torsion, patellar alignment and any signs of muscle wasting
(b) Observation—supine. Observe for lower limb alignment, effusion, position of the patella, and any evidence of patella tilt or rotation (c) Functional test—double/single-leg squat
If the patient's pain has not already been reproduced, functional tests such as squat, lunge, hop, step-up, step-down or eccentric drop squat should be performed



(b)



(c)



(d)



(f)



(e)

Figure 36.2 (cont.) (d) Functional test—single-leg squat on decline board (e) Palpation—patella, and medial and lateral facets are palpated for tenderness (f) Patellofemoral joint assessment—mobility of the patella—medial glide

3. Palpation

- a. patella—medial and lateral facets (Fig. 36.2e)
- b. medial/lateral retinaculum
- c. patellar tendon
- d. tibial tubercle
- e. effusion—swipe test, ballotment test

4. PFJ assessment

- a. quadriceps function
- b. mobility of patella
 - i. superior glide
 - ii. inferior glide
 - iii. medial glide (Fig. 36.2f)
 - iv. lateral glide—look for apprehension
- c. tracking of patella in passive knee flexion/extension

5. Flexibility and range of motion

- a. lateral soft tissue structures
- b. quadriceps
- c. hamstring
- d. iliotibial band
- e. gastrocnemius
- f. soleus

Table 36.3 Diagnostic accuracy of examination tests for patellofemoral pain

Test	What is a positive finding?	Sensitivity	Specificity	LR+	LR–
Pain during functional tests (history or examination)					
Squatting ¹⁰	Pain	91	50	1.8	0.2
Stair climbing ¹⁰	Pain	72	43	1.3	0.6
Kneeling ¹⁰	Pain	84	50	1.7	0.3
Prolonged sitting ¹⁰	Pain	72	57	1.7	0.5
Patellar palpation					
Medial tenderness ¹¹	Pain	48	71	1.6	0.7
Lateral tenderness ¹¹	Pain	41	71	1.4	0.8
Palpation of patellar borders ¹⁰	Pain	47	68	1.5	0.8
Passive mobility tests					
Passive gliding patella ¹¹	Pain	48	47	0.9	1.1
Patella translation superior to inferior ¹²	Reduced ROM	63	56	1.4	0.7
Patella translation medial to lateral ¹²	Reduced ROM	54	69	1.8	0.7
Patellar tendon mobility ¹²	Reduced ROM	49	83	2.8	0.6
Patella inferior pole tilt ¹²	Reduced ROM	19	83	1.1	0.9
Special tests					
Patella apprehension test ¹³	Pain or apprehension	15	89	1.3	1.0
Resisted isometric quads ¹⁰	Pain	39	82	2.2	0.8

LR+ = positive likelihood ratio; LR– = negative likelihood ratio

See Chapter 14 for a guide to interpreting test sensitivity, specificity and likelihood ratios

Table 36.4 Diagnostic accuracy of examination tests for patellar tendinopathy

Test	What is a positive finding?	Sensitivity	Specificity	LR+	LR–
Palpation					
Palpation of the patellar tendon attachment ⁹	Pain	68	9	0.5	0.2

LR+ = positive likelihood ratio; LR– = negative likelihood ratio; NA = not available

See Chapter 14 for a guide to interpreting test sensitivity, specificity and likelihood ratios

- g. knee flexion/extension
- h. tibial rotation
- i. hip and lumbar spine range of motion—all planes
6. Special tests (to exclude other pathology)
 - a. examination of knee joint with respect to acute injury (Chapter 35)
 - b. examination of hip joint (Chapter 31)
 - c. examination of lumbar spine (Chapter 29)
 - d. neurodynamic tests (neural Thomas test, slump test, prone bent knee test-Chapter 9)

Patient-reported outcome measures

Patient reported outcome measures (PROMs) are considered gold standard to evaluate outcomes in a clinical population. When choosing a PROM, the psychometric properties of all possible PROMs should be considered, specific to the patient. For PFP, three scales (the anterior knee pain scale, and visual analogue scales for usual and worst pain) have adequate psychometric properties for use (Table 36.5). The patient specific functional scale¹⁴ is also a useful clinical PROM, but has not been



Table 36.5 Patient-reported outcome measures for patellofemoral pain

Anterior knee pain scale (AKPS), also known as Kujala scale
• Self-administered questionnaire that evaluates pain and function in anterior knee pain¹⁵ • 13 items • Score range 0–100 (100 is best score) • Test-retest reliability: 0.81–0.90¹⁶ • MDC95%: 19.7¹⁶–9.7¹⁷ • MCID: 8–10¹⁷
Visual analogue pain scale when usual pain (VAS-U)
• Self-administered questionnaire to evaluate pain on a 100 mm visual analogue scale during usual activities in the past week¹⁷ • Score range 0–100 (100 is worst imaginable pain) • Test-retest reliability: 0.56–0.77¹⁶ • MDC95%: 33¹⁶–17¹⁷ • MCID: 15–20¹⁷
Visual analogue pain scale when worst pain (VAS-W)
• Self-administered questionnaire to evaluate pain on a 100 mm visual analogue scale during worst (most aggravating) activities in the past week¹⁷ • Score range 0–100 (100 is worst imaginable pain) • Test-retest reliability: 0.56–0.79¹⁶ • MDC95%: 30¹⁶–17¹⁷ • MCID: 20¹⁷

specifically validated for PFP. For each scale, the patient needs to improve by more than the Minimal Detectable Change (MDC), to be greater than the error associated with the measurement and greater than the Minimal Clinically Important Difference (MCID), to have a clinically meaningful change. See Chapter 16 for more details on PROMS and how to use them in clinical practice.

Investigations

Imaging may be used to confirm a clinical impression obtained from the history and examination. Structural imaging includes plain radiography, ultrasound, CT, and MRI. Occasionally, radionuclide bone scan is indicated to evaluate the ‘metabolic’ status of the knee (e.g. after trauma, or in suspected stress fracture).

Most patients with PFP will require either no imaging or plain radiography consisting of a standard AP view, a true lateral view with the knee in 30° of flexion and an axial view through the knee in 30–40° of flexion. Plain radiography can detect bipartite patella, apophyseal changes at the patellar tendon attachments and osteoarthritis, as well as rule out potentially serious complications such as tumour or infection. Although CT and three-dimensional CT can assess the PFJ morphology, MRI is gaining increasing popularity as an investigation of

PFP and the unstable patella.¹⁸ These imaging techniques can identify features including bone lesions, chondral lesions, tendinopathy and synovitis. However, the importance of such features to symptoms or progression of disease remains unknown.¹⁹

MRI displays high signal abnormality in patellar tendinopathy. Ultrasound hypoechogenicity and excessive vascularity also indicate a potential diagnosis of patellar tendinopathy.²⁰ However, in asymptomatic athletes or those with symptoms from another source in the anterior knee, the patellar tendon can be abnormal on imaging,^{21, 22, 23} and the clinician should be wary of using imaging findings diagnostically.

PATELLOFEMORAL PAIN

In this section we:

- define the condition and underscore its significance
- review clinically relevant functional anatomy
- alert the clinician to predisposing factors and how to identify them—this may prove critical for effective long-term treatment
- summarise the results of RCTs that have added high-quality evidence (Chapter 2) to guide clinical treatment.

What is patellofemoral pain?

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Patellofemoral pain is the preferred term used to describe pain in and around the patella.

Synonyms include patellofemoral pain syndrome, anterior knee pain and chondromalacia patellae. Patellofemoral pain (PFP) is an 'umbrella' term used to embrace all peripatellar or retropatellar pain in the absence of other pathologies.²⁴ Since the cause of the pain may differ between patients, it is appropriate to review the potential sources of PFP.²⁵

Numerous structures in the PFJ are susceptible to overload. A number of extra- and intra-articular components of the knee can generate neurosensory signals that ultimately result in the patient feeling pain (see box

below). Patellofemoral articular cartilage is avascular and aneural. However, a cartilage lesion may lead to chemical or mechanical synovial irritation, bone oedema or erosion, all of which can result in pain.^{26, 27}

Peripatellar synovitis, in the absence of obvious cartilage damage, must be considered as a primary cause of PFP.²⁸ Soft tissues such as the lateral retinaculum are also implicated as a potent source of PFP.^{29, 30-32} Other important sources of pain are the synovium and infrapatellar fat pad.³³⁻³⁶

So how does an increase in PFJ load result in PFP? Dye and Vaupel³⁷ described a concept whereby injury to the PFJ musculoskeletal tissues results from supraphysiological loads—either a single maximal load or lower magnitude repetitive load. Injury to these tissues initiates a cascade of events encompassing inflammation of the peripatellar synovium through to bone stress. Thus, a number of different pain-sensitive structures can give rise to the conscious sensation of PFP (Chapters 5 and 6).³⁷

The 'pain' in patellofemoral pain

Local and central contributors to PFP

Pain is a complex protective mechanism (Chapters 5 and 6). Although nociception is an input to the brain that may result in the output of pain, the brain considers many other inputs before deciding to create pain, some of which may be far more decisive than the level of nociception. The experience of knee pain is modulated at different sites in the body, both locally and centrally within the brain and spinal cord. This means that both local peripheral nociceptors and facilitated central pain pathways, may influence the pain that is experienced by the patient. Current thinking in the field of pain science teaches us that pain reflects the perception of threat more so than the state of injury to tissues. Pain, such as perceived in the PFJ, is considered a driver of behaviour to protect tissues.

Do people with PFP have hyperalgesia?

The pathways involved in PFP are starting to be evaluated. Compared with pain-free controls, female adolescents presenting with longstanding PFP have hypersensitivity highlighted by reduced pressure pain thresholds (PPTs) around the knee and more distally around the tibialis anterior muscle.³⁸ The distal hyperalgesia may reflect segmental spreading of hyperalgesia.³⁹ Identification of pressure pain hyperalgesia (decreased PPT) may be clinically important and associated with poorer recovery as previously shown in adult patients with acute whiplash injury and after total knee replacement. However, early reports indicate that low PPTs around the knee or tibialis anterior are not associated with

a worse prognosis and PPTs tend to increase when patients recover.⁴⁰ This implies a nociceptive contribution to PFP.

How does facilitated central pain mechanism affect the pain experience in PFP?

Conditioned pain modulation (CPM) (primarily a descending inhibitory mechanism) is also impaired in young female adults with longstanding PFP.⁴⁰ Less efficient CPM mechanisms are evident in patients with musculoskeletal pain conditions, such as myofascial temporomandibular disorders, chronic low back pain and fibromyalgia. A reduced potency of the descending control can make the entire neuroaxis more vulnerable to pain. However, in the study of female adolescents with PFP, the CPM mechanisms were highly variable, ranging from an efficient CPM system to severely impaired CPM. This suggests that central pain mechanisms may be a factor we need to consider in some, but not all, patients with PFP. Furthermore, there is emerging evidence that some people may be more vulnerable to pain due to genetic and other factors.

Clinical signs that may suggest facilitated central pain mechanisms

In a clinical setting the following signs may suggest facilitated central pain mechanisms:

- disproportionate pain, implying that the severity of pain and related reported or perceived disability are disproportionate to the nature and extent of the injury
- presence of diffuse pain distribution, allodynia and hyperalgesia.

If this fits with your patient with longstanding PFP and he or she does not respond to your normal exercise protocol, you might want to consider the following items.

Guidelines for clinical management of longstanding PFP in respect to a heightened/hypersensitive central pain presentation
(adapted from Nijs et al⁴¹)

- Be conservative when estimating the initial training load; prefer a lower baseline and then progress from there depending on symptom flaring.

- Monitor symptom flares after exercise and progress exercises depending on response.
- Let the patient know that minor symptom flares are natural during the initial stages of exercise therapy, but should cease once an exercise routine is established.
- Allow increased pain during and shortly following exercise but avoid continuously increasing pain intensity over time.
- Use aerobic exercise as well as motor control training.
- Include exercise of non-painful parts of the body (including cross-education and strength training of the other limb).
- Use multiple and long recovery breaks in between exercises.

What is patellofemoral osteoarthritis?

Patellofemoral osteoarthritis (OA) can be a considerable source of anterior knee symptoms and functional limitations,⁴² with a growing body of evidence indicating that patellofemoral OA occurs before tibiofemoral disease⁴³ and can affect young to middle-aged adults.⁴⁴

Specific risk factors for early-onset patellofemoral OA development include a history of patellofemoral dysfunction (i.e. pain, dislocation, cartilage lesions), ACL rupture, and meniscal tears. Evidence updates, to highlight the importance of patellofemoral OA, are outlined in the box below.

Patellofemoral osteoarthritis as a source of anterior knee pain in active young adults

Is patellofemoral OA related to PFP?

The best available evidence suggests that PFP and patellofemoral OA may exist on a continuum;⁴⁵ retrospective recall of PFP is associated with radiographic⁴⁶ and MRI⁴⁷ features of patellofemoral OA. While not all patients with PFP will develop patellofemoral OA, PFP should be considered to be more than simply a benign, self-limiting condition and therefore be managed with appropriate evidence-based interventions.

Is patellofemoral OA a consequence of acute knee injuries?

Anterior cruciate ligament injury increases the risk of tibiofemoral and patellofemoral OA.⁴⁸ Five years after ACL injury, radiographic patellofemoral OA is evident in 19% of knees, regardless of conservative or surgical management or graft type used,⁴⁹ while MRI evidence of patellofemoral OA is observed in 17% of individuals at 1 year following ACL reconstruction.⁵⁰ By 10 years after ACL reconstruction, almost 50% of people will demonstrate radiographic patellofemoral OA.⁵¹ Meniscal tears requiring partial meniscectomy, either in association with ACLR⁵² or in isolation,⁵³ are associated with 2–5 times greater odds of patellofemoral OA development than knees without meniscal pathology over the following 8–20 years.

The mechanisms driving the development of post-traumatic patellofemoral OA following ACL and meniscal injury are not well understood, but may relate to concomitant damage to the PFJ at the time of initial trauma, or inherent

biomechanics that are risk factors for both PFP and ACL injury (e.g. high knee abduction moments).⁵⁴

Patellofemoral trauma, such as patellar dislocation, doubles the risk of patellofemoral OA at 13 years post-injury (22% vs 11%), irrespective of non-operative or surgical management.⁵⁵

The onset of post-traumatic patellofemoral OA may be mediated by the presence of a patellofemoral articular cartilage lesion—a particularly common finding in the unstable PFJ (97%),⁵⁶ which predicts future cartilage loss⁵⁷ and the development of radiographic patellofemoral OA.⁵¹

Can we identify people with or at high risk for patellofemoral OA?

The presence of knee crepitus and history of patellar pain are associated with patellofemoral joint OA (but not tibiofemoral joint OA) in women.⁵⁸ However, clinical examination findings and knee pain location have limited ability to discriminate those with patellofemoral OA from those with tibiofemoral OA.²⁴ Future studies may shed some light on diagnostic tests to recognise or predict patellofemoral OA in people with PFP.

Where to now for research and clinical practice?

The sensitivity of MRI to detect joint changes in the earliest stages of disease and recently proposed MRI-OA criteria⁵⁹ offer promise for early OA diagnosis, enabling studies of risk factors for early onset patellofemoral OA and evaluating interventions to be tested early in the disease process.

Functional anatomy

At full extension, the patella sits lateral to the trochlea. During flexion, the patella moves medially and comes to lie within the intercondylar notch until 130° of flexion, when it starts to move laterally again.³⁵ With increasing knee flexion, a greater area of patellar articular surface comes into contact with the femur, thus offsetting the increased contact load that occurs with flexion. Loaded knee flexion activities subject the PFJ to loads ranging from 0.5 times body weight for level walking to seven to eight times body weight for stair climbing.³⁶

Factors that may contribute to patellofemoral pain

Patellofemoral pain is likely to be initiated by increased or unaccustomed PFJ loads. Factors that influence PFJ load can be considered in two categories: extrinsic and intrinsic. During physical activities the extrinsic load is created by the body's contact with the ground (ground reaction force) and is therefore moderated by body mass, speed of gait, surfaces and footwear. The number of loading cycles and frequency of loading are also important. During weight-bearing activities, any increase in the amount of knee flexion will increase the PFJ load. Therefore, when an individual experiences an increase in the magnitude of the PFJ load (e.g. higher training volume, increased speed of running), this may overload the PFJ structures sufficiently to initiate pain. Note that activities with a high eccentric component, such as bounding, will overload the PFJ structures but are more likely to lead to patellar tendon pain.

Intrinsic factors can influence both the magnitude and the distribution of the PFJ load. Distribution of load is influenced by the alignment of the patella within the femoral trochlea, also known as patella tracking. Intrinsic factors that can influence patella tracking may be considered as 'remote' or 'local'. Considering the patient as a whole, 'remote' factors that affect patella tracking include femoral and tibial rotations. These factors can in turn be influenced by hip muscle strength, trunk kinematics, subtalar pronation and muscle flexibility (discussed below).⁶⁰ Local factors that influence patella movement include bony shape of the patella and trochlea, soft tissue tension and neuromuscular control of the medial and lateral components of the vasti.⁶¹

Therefore, when the initial subjective and objective examinations are completed and the diagnosis of PFP is confirmed, the clinician should assess the contribution of various extrinsic and intrinsic factors to the development of PFP.

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Although the history provides valuable information pertaining to extrinsic factors, clinical examination is usually required to evaluate the key intrinsic contributing factors ('remote' and 'local'). This comprehensive assessment is crucial in the planning treatment.

'Remote' intrinsic factors

The following 'remote' intrinsic factors may contribute to PFP developing (see also Chapter 8):

- increased hip/femoral internal rotation
- Increased hip adduction
- increased apparent knee valgus/external tibial rotation
- poor trunk and pelvic control
- pronated foot type
- sagittal plane motions.

It is important to assess the patient in static postures and during functional activities. Some factors may become more obvious during specific functional tasks, such as the step-down or single-leg squat and hopping or landing from a jump, where the postural demands are high. Once a potential contributing factor has been identified, the clinician must investigate the mechanisms that may require intervention (Table 36.6a-d).

Increased femoral/hip internal rotation

Increased femoral/hip internal rotation motions are associated with PFP⁶²⁻⁶⁵ and may contribute to its development.⁶⁶ Clinical observation of the patient in standing reveals internally rotated femurs, often manifesting as 'squinting patellae'—the patellae both face medially. During tasks of increasing load, further internal rotation of the thigh can often be observed and may be visualised as an apparent knee valgus.

Increased hip adduction motions

Increased hip adduction motion may increase valgus load on the knee during dynamic tasks⁶² and has been associated with PFP.⁶³⁻⁶⁸ Clinical observation of the patient in standing may not reveal any abnormality. During functional tasks such as a single-leg squat, adduction of the thigh can often be observed and is usually associated with a pelvic drop towards the non weight-bearing leg.

Increased apparent knee valgus and/or increased tibial external rotation

Genu valgum (knee valgus or abduction) may be associated with PFP and observed in standing. This

Table 36.6a Remote factors that can contribute to patellofemoral pain syndrome and their possible mechanisms—increased femoral internal rotation

Increased femoral/hip rotation	Possible contributing factors	Confirmatory assessments
	Structural: • femoral anteversion	Imaging—MRI, radiography
	Inadequate strength/control: • hip external rotators • hip abductors • hip extensors • lumbopelvic muscles	Clinical assessment: • manual muscle test (Figs 28.3a, b) • hand-held dynamometer • biofeedback
	Range of motion deficits: • inadequate ankle dorsiflexion range • inadequate hip external rotation range	Range of motion tests: • clinical inclinometer (Chapter 8) • figure '4' or FABER test (Fig. 31.6c)
	Altered movement patterns	Biomechanical/gait assessment (Chapter 8): • can the patient correct the movement pattern when asked to focus on it?

Table 36.6b Remote factors that can contribute to patellofemoral pain syndrome and their possible mechanisms—increased hip adduction

Increased hip adduction	Possible contributing factors	Confirmatory assessments
	Inadequate strength/control: • hip external rotators • hip abductors • hip extensors • lumbopelvic muscles • trunk muscles	Clinical assessment: • manual muscle test • hand-held dynamometer • side-bridge tests • biofeedback
	Altered movement patterns	Biomechanical/gait assessment (Chapter 8): • can the patient correct the movement pattern when asked to focus on it?

may be exaggerated during gait, possibly associated with a midstance valgus thrust. Additionally, increased hip adduction/internal rotation or a lateral pelvic drop during the step-down and single-leg squats (potentially owing to weakness of the gluteus medius) may be observed as an increase in apparent knee valgus posture. Knee abduction is associated with PFP.^{63,65, 67, 69, 70} Increased structural or functional tibial rotations can

affect PFJ loads directly and also through transferred rotations to the femur. Importantly, tibial rotations are strongly coupled with the motion of the subtalar joint. Although there is little data on the assessment or treatment of structural tibial rotation in isolation, experienced clinicians often address functional rotation in association with femoral or subtalar rotations (see treatment outline later).

Table 36.6c Remote factors that can contribute to patellofemoral pain syndrome and their possible mechanisms—increased knee valgus/tibial external rotation

Increased apparent knee valgus/tibial external rotation	Possible contributing factors	Confirmatory assessments
	Structural: • genu varum • tibial varum • coxa varum	Radiographic: • long leg X-ray Clinical assessment: • goniometer/inclinometer
	Inadequate strength/control: • hip external rotators • hip extensors • hip abductors • quadriceps • hamstrings • lumbopelvic muscles	Clinical assessment: • manual muscle test • hand-held dynamometer • biofeedback
	Altered movement patterns	Biomechanical/gait assessment (Chapter 8): • can the patient correct the movement pattern when asked to focus on it?

Table 36.6d Remote factors that can contribute to patellofemoral pain syndrome and their possible mechanisms—increased trunk lean

Increased trunk lean	Possible contributing factors	Confirmatory assessments
	Inadequate strength/control: • trunk extensors/flexors/rotators • hip abductors/adductors • hip extensors/flexors • lumbopelvic muscles	Clinical assessment: • manual muscle test • hand-held dynamometer • biofeedback
	Inadequate strength/control: • hip external rotators • hip extensors • hip abductors • quadriceps • hamstrings • lumbopelvic muscles	Clinical assessment: • manual muscle test • hand-held dynamometer • biofeedback
	Altered movement patterns	Biomechanical/gait assessment (Chapter 8): • can the patient correct the movement pattern when asked to focus on it?

Trunk and pelvic motions

The trunk represents the largest proportion of body mass and hence any alterations to trunk and pelvic motions can influence lower limb loads, including those of the PFJ. Although not studied as frequently as other segments, recent studies have focused on trunk and pelvic motions and their influence on PFJ loads.⁷¹ Increased trunk

flexion is associated with lower PFJ stress.⁷¹ Furthermore, people with PFP exhibited greater anterior (forward) and ipsilateral trunk lean and contralateral trunk drop, compared to controls.^{63, 65, 72, 73} Clinical observation of the patient in standing may not reveal any abnormality. During functional activities such as a single leg squat, ipsilateral trunk lean may be observed, often in association

with contralateral pelvic drop. Observing the patient from the side may reveal inadequacies in sagittal plane motion (i.e. too much or too little forward trunk lean or anterior pelvic tilt).

Dynamic foot function

Dynamic foot function (e.g. plantar loading) and static foot posture (e.g. increased pronation, navicular drop, arch height mobility) are associated with PFP and may contribute to its development.^{63, 65, 66, 74-78} Static foot posture can be observed clinically in standing and during gait (Chapter 8).

Sagittal plane

Recently, studies have focused on frontal plane and transverse plane motions when evaluating people with PFP. Sagittal plane impairments should also be considered. In particular, loss of knee flexion⁷⁹⁻⁸¹ and greater trunk flexion^{72, 73, 82} are

features of people with PFP. While some of these adaptations may serve to reduce PFJ reaction force and hence joint stress, over time they may be associated with atrophy of key muscle groups (especially quadriceps) or increased frontal plane knee loads⁸³ with potential for further damage to the PFJ.⁸⁴ Hence, all patients should be observed from the side, to check for loss of sagittal plane motions.

Local contributing factors

Local factors that can contribute to the development of PFP are in three main categories (Table 36.7):

- patella position
- soft tissue contributions
- quadriceps (vasti) dysfunction.

Patella position

Clinical examination provides valuable information on the structural and functional relationships of the PFJ

Table 36.7 Local factors that can contribute to patellofemoral pain

Factor	Structural observations	Functional observations (with quadriceps contraction)
Patella position		
Lateral displacement	• Patella displaced laterally, closer to the lateral than medial femoral condyle • Restricted medial glide (Fig. 36.2f)	• Patella moves laterally
Lateral tilt	• Difficult to palpate lateral border, high medial border • Lateral tilt increases with passive medial glide	• Patella tilts laterally
Posterior tilt	• Inferior patella pole displaced posteriorly, often difficult to palpate due to infrapatellar fat pad	• Inferior pole moves further posteriorly. A 'dimple' may appear in the infrapatellar fat pad
Patella alta	• High-riding patella	• NA
Soft tissue contributions		
Tight lateral structures	• Lateral patella displacement or tilt • Palpation of lateral structures (Fig. 36.3)	• Lateral patella displacement or tilt
Compliant medial structures	• Lateral patella displacement or tilt	• Lateral patella displacement or tilt
General hypermobility	• NA	• Increased patellar mobility in all directions
Quadriceps (vasti) dysfunction		
Reduced activity of quadriceps (general)	• Reduced muscle bulk of quadriceps	• Reduced muscle strength
Delayed onset or reduced magnitude of VMO relative to VL	• Reduced muscle bulk of VMO	• Poor quantity/quality of VMO • Assess in functional positions • Biofeedback can assist (Fig. 36.4)

(Table 36.7). The clinician should carefully assess passive and active movement of the patella in all directions: medial (Fig. 36.2f), lateral, superior, and inferior glides, as well as rotations. Although the tests for patella position are not functional and may not be repeatable,^{85, 86} clinical examination of the patella position remains a useful tool for clinical decision making.

Soft tissue contributions

The contribution of the superficial and deep soft tissues to the PFJ mechanics can, in part, be obtained from the structural and functional assessments of patella position. Palpation around the thigh, knee and lower leg allows the clinician to ascertain the compliance of soft tissues which may be contributing to PFP (Fig. 36.3).

Quadriceps (vasti) dysfunction

Individuals with PFP produce less quadriceps torque⁸⁷⁻⁸⁹ and have smaller quadriceps size⁹⁰ than those without pain, and quadriceps weakness increases the risk of PFP development.⁹¹

There is controversy in the literature regarding an imbalance in vasti activation in PFP.⁹²⁻¹⁰¹ Inadequate vasti motor control could increase lateral patellar shift¹⁰² or patellofemoral pressure.¹⁰³ Balanced activation of VMO and VL has long been believed to be disrupted in people with PFP, but the literature is inconsistent regarding imbalances in vasti activation or size and PFP pain.^{99-101, 104-107} These inconsistencies may in part be accounted for by a wide variation in methodology and also by inherent heterogeneity in the PFP population, meaning that some, but not all, may have delayed VMO onset. Furthermore, the role of delayed EMG onset on medial vasti force production remains unknown.

Clinical examination of the vasti (medial and lateral components) provides some insight into their role in a patient's presentation (Table 36.7, Fig. 36.4). Frank muscle wasting and weakness may be obvious, but seemingly normal muscle bulk does not ensure normal function. If VMO dysfunction is suspected, then its ability to contract synchronously with the rest of the quadriceps should be evaluated. The vasti function should be assessed in a number of positions or activities, including those that are functionally relevant to the patient.

Treatment of PFP

Effective treatments usually integrate several different treatment techniques that target the local PFJ factors, in addition to the more remote (distal and proximal) factors. Different techniques, including exercise, manual therapy, taping, bracing and foot orthoses can be used in isolation or in combination, depending on the patient's presentation, the clinician's expertise and the preferences



Figure 36.3 Clinical assessment of lateral soft tissue contribution. The iliotibial band and lateral retinacular structures are palpated for any reduced compliance

of both patient and clinician.¹⁰⁸ An integrated approach to management of a patient with PFP generally includes the following components:

- exercise
- taping
- manual therapy
- foot orthoses
- bracing.

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The first priority of treatment is to reduce pain. Rest from aggravating activity usually suffices but it may require ice, a short course of oral analgesics (rarely) or techniques such as mobilisation (Fig. 36.7). Taping should have an immediate pain-relieving effect; when it does, this strongly suggests that the diagnosis is PFP (Fig. 36.6).

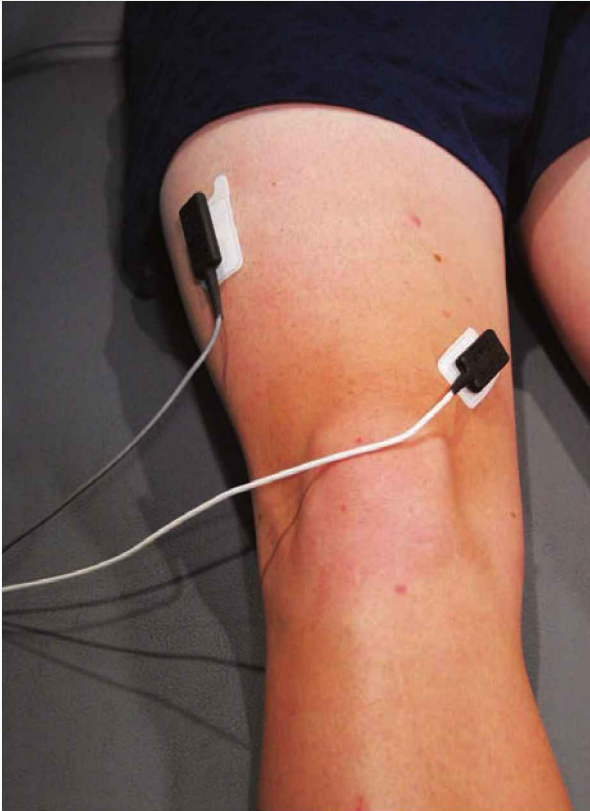


Figure 36.4 Clinical examination of neuromotor control of the vasti. The quality (amount and timing) of the vastus medialis obliquus and vastus lateralis component of the quadriceps can be assessed in a number of positions, including supine, sitting and standing. In these positions the patient is asked to contract the quadriceps. The clinician can either observe or palpate the quality of the contraction in these positions. A surface EMG biofeedback (illustrated) may be used to provide useful information to the clinician on the relative contribution of the vasti during quadriceps contractions across the various positions

Exercise

Exercise is a cornerstone of PFP management. Usual components include:

- quadriceps retraining
- hip muscle retraining
- strengthening/endurance building for lower limb and trunk muscles
- coordination and balance training
- retraining of functional activities (including sports- or work-related).

Not all patients will require all components. Using a pain monitoring system¹⁰⁹ (where pain levels scoring 0-3

on a 10 cm numerical pain scale are considered acceptable during exercise) may avoid pain-induced muscle inhibition and enhance adherence to an exercise program. Pain levels up to 5 (out of 10), momentarily during exercise or immediate following exercise are acceptable, but not extending to the following morning. Patellar taping may be required to ensure pain-free exercise.

Quadriceps retraining programs

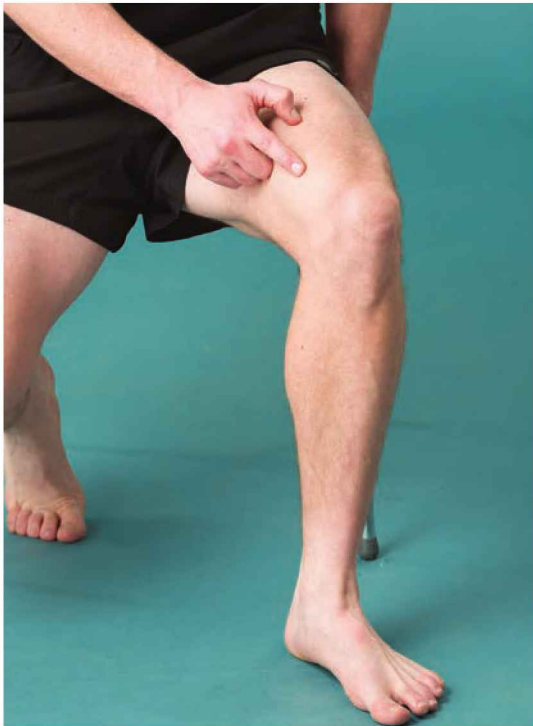
Retraining the quadriceps can reduce PFP and symptoms,^{108, 110-113} as well as enhance VMO activation relative to VL.¹¹⁴ Patients cannot activate the VMO in isolation and no exercise appears to be preferential for activation of the VMO. Most people will not require a specific focus or emphasis on their VMO. However, for those who appear to have difficulty with VMO activation, it may be necessary to find and use training positions where the patient can attain a consistent VMO activation.

It may be necessary for the patient to learn to contract the VMO. By palpating the VMO while contracting their quadriceps in various degrees of knee flexion, hip abduction or knee rotation and/or in different activities the patient can receive feedback on the contraction. A dual-channel biofeedback machine may also be used (Fig. 36.4). The patient should attempt to recruit the VMO during all quadriceps contractions. Other strategies may assist VMO activation, including imagery or visual cues, facilitation techniques (e.g. palpation, dry needles/acupuncture), other forms of feedback (e.g. real-time ultrasound, tactile or visual) or inhibitory techniques to reduce over-activity in the VL or lateral hamstrings (e.g. inhibitory taping, feedback).

Initially, quadriceps exercises may commence in sitting with the knee at 90°, the foot on the floor and in this position, the patient may palpate the VMO to facilitate muscle activation (Fig. 36.5a). To ensure the quadriceps are trained in positions where they are required to function, the patient should begin training in a weight-bearing position and perform functional exercises with steadily increasing load and difficulty as soon as possible.

Progression is based on their ability to maintain control and the absence of significant pain (2-3 out of 10). Lower load exercises may include lunges, squats and step-ups. Further progressions can be made from minimal weight-bearing (using hand rails or other supports) to full weight-bearing (eventually including added loading) and through various knee flexion ranges.

The final aim of quadriceps training is to achieve a carry-over from functional exercises to functional activities. The patient should perform small numbers of exercises frequently throughout the day. A series of graded quadriceps exercises is demonstrated in Figure 36.5.



(a)



(c)



(b)



(d)

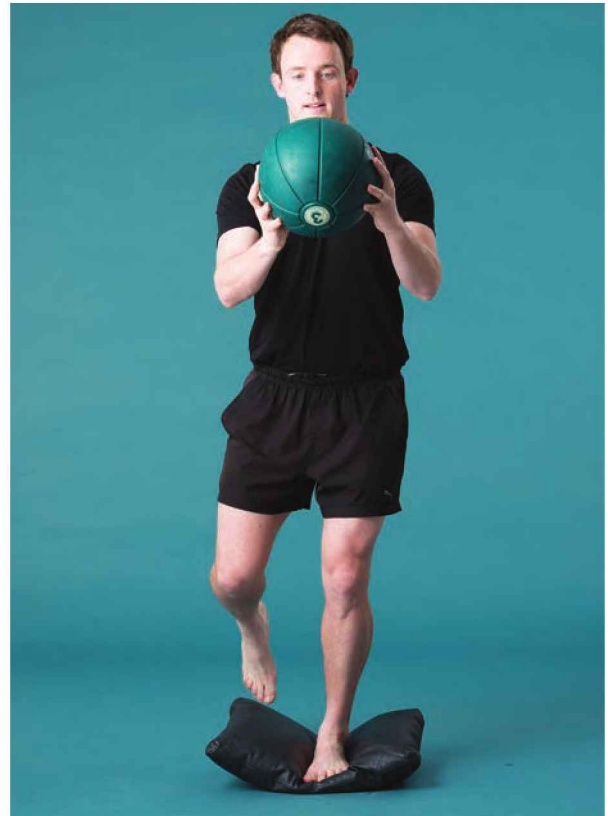
Figure 36.5 (a) Retraining focuses on increasing the patient's ability to activate their VMO, ideally with little VL activity or lateral hamstring co-contraction (b) Step-up with feedback on knee position (c) Double-leg squat (d) Step-up



(e)



(f)

Figure 36.5 (cont.) (e) Single-leg step-down (f) Lunge

(g)

Figure 36.5 (cont.) (g) Standing on one leg, with single-leg squat catching a ball

Hip retraining programs

Retraining the hip abductors, external rotators and extensors is thought to stabilise the lateral pelvis and to control internal hip rotation and adduction. Such strengthening programs have been associated with pain reduction in patients with PFP.¹¹⁵ Indeed, evidence suggests that hip retraining is an essential component to any PFP rehabilitation. The principles for retraining hip muscle function are the same as above, with exercises performed initially in non weight-bearing positions and then progressed to weight-bearing positions (Fig. 36.5). It is critical to ensure that the targeted muscles are being activated during the retraining program. For example, if your assessment reveals that hip external rotator retraining is required then the retraining programs must be activating the hip external rotators. More detail on hip retraining exercises is available in Chapter 31. The focus of all retraining exercises is on quality, then quantity.

As soon as it is possible and practical, the patient must be taught to activate the hip abductors, extensors and

external rotators, in combination with other lower limb muscles during functional exercises. The emphasis of all retraining exercises will be on maintaining the activation of these muscles and correct alignment of the hip (i.e. neutral rotation) during weight-bearing flexion tasks (e.g. lunge, step up and step down). Some patients may require retraining of their movement patterns during functional or potentially aggravating activities.¹¹⁶

Lower limb strengthening/endurance-building exercises

Adequate strength and endurance, especially in the quadriceps, hip, trunk and calf muscles, is likely to be important for all patients. Strengthening programs can be commenced early (see above for maintaining control and monitoring pain) in the patient's treatment, and progressed according to patient response and American College of Sports Medicine guidelines.¹¹⁷ A generalised quadriceps strengthening program can effectively relieve patellofemoral pain and improve function.¹¹² However,

patients with poor vasti or hip coordination may first need to address their motor control deficits. The available evidence does not support the effectiveness of one exercise regimen over another.

Movement retraining

This is required in the final stage of the exercise-based interventions in order to return to high loaded activities involving knee flexion during full weight-bearing (e.g. stair descent, deep squats), or higher intensity activities such as running. Patients need additional training to facilitate progression to these high PFJ loading activities. Motor control, strength and endurance in the quadriceps and global muscles (e.g. calf, hip and trunk muscles), balance and coordination should be trained in these higher loading tasks. Exercise choice and progression decisions are based on the patient's needs, their ability to maintain control and the absence of significant pain. More detail on movement (and running) retraining is available in Chapter 8.

Patellar taping

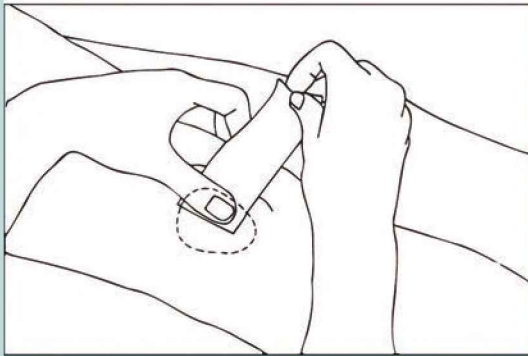
The aim of taping is to correct the abnormal position of the patella in relation to the femur. Although patellar taping reduces PFP substantially and immediately,¹¹⁸ the precise mechanism of the effect is still being investigated. It is possible that pain relief following patella taping could be due to subtle changes in patella position and PFJ contact area¹¹⁹ and/or changes in motor control or proprioceptive input.¹²⁰ Regardless of the mechanism, patella taping is an effective interim measure to relieve PFP and can increase adherence to an exercise program and/or enable pain-free activities of daily living.

A commonly used technique involves taping the patella with a medial glide (Fig. 36.6a). It may also require correction of abnormal lateral tilt (Fig. 36.6b), rotation (Fig. 36.6c) or inferior tilt (Fig. 36.6d). The taping is performed with rigid strapping tape. It is important that the clinician recognises a posteriorly displaced inferior pole of the patella, as taping the patella too low will increase the patient's symptoms. Taping to 'unload' the fat pad (effectively to 'lift' the patella away from

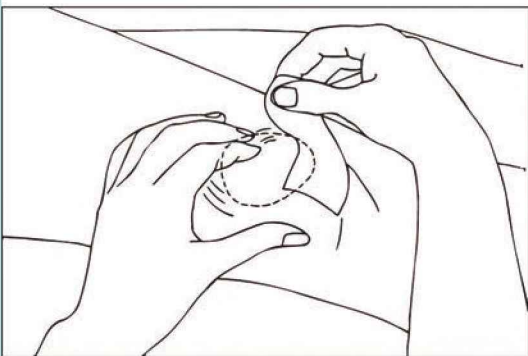


(a)

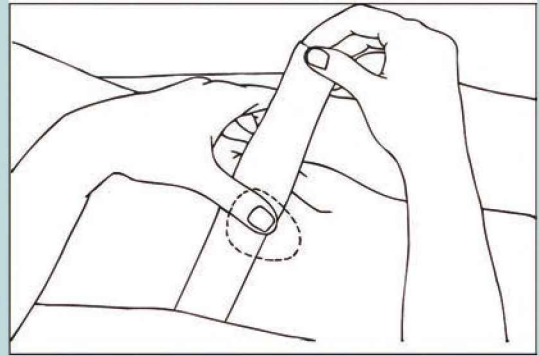
Figure 36.6 Patellar taping techniques (a) Patellar taping (medial glide). Tape is applied to the lateral aspect of the patella. The patella is glided medially and the tape is anchored to the skin over the medial aspect of the knee. When taping is completed, skin creases should be evident on the inside of the knee, indicating adequate tension on the patella



(b)



(c)



(d)

Figure 36.6 (cont.) (b) Patellar taping (correction of lateral tilt). Tape is applied to the medial aspect of the patella and secured to the soft tissue on the inner aspect of the knee (c) Patellar taping (correction of rotation). Tape is applied to the inferior pole of the patella and taken medially and superiorly to rotate the patella (d) Patellar taping (correction of inferior tilt). Tape is applied across the superior pole of the patella with sufficient firmness to elevate the inferior pole

the fat pad) can also be used, either alone or in conjunction with the above techniques (see below, 'Fat pad irritation/impingement' for more details; Fig. 36.15).

Patella taping effects should be assessed immediately using a pain-provoking activity such as a single- or double-leg squat. If the tape has been applied correctly and is going to be a useful treatment option, the post-taping squat will be less painful. If all or some pain persists, the tape should be altered, possibly including a component for tilt or rotation or both. Note that if patella tape does not make a substantial (at least 50%) reduction in the patient's pain, the initial diagnosis should be revisited.

If patients are able to perform strengthening exercises pain-free without tape, then exercises should be performed

continued

tape free. Some people, however, require tape to perform the exercises and, initially, to continue their sporting activities. Acute cases of PFP may initially need tape applied 24 hours a day until the condition settles. The tape time is then gradually reduced.

Adverse skin reactions can occur beneath the rigid tape. Therefore, the area to be taped should be shaved and a protective barrier applied beneath the rigid strapping tape to reduce both the reaction to the zinc oxide in the tape

adhesive and the reaction to shearing stresses on the skin. This can be achieved with adhesive gauze tape (Hypafix® or Fixomull®) applied to the area to be taped. A protective barrier or plastic skin can also be used in patients with extremely sensitive skin. If skin irritation still occurs, the patient must be advised to remove the tape. Treatment with a hydrocortisone cream may be necessary. Patients with fair skin seem to have particularly sensitive skin and need to be monitored closely.

Manual therapies

Manual therapies, such as joint and soft tissue mobilisation, manipulation and massage may relieve symptoms associated with PFP. Joint mobilisations can include patellofemoral or tibiofemoral joint glides, whereas soft tissue therapies might involve massage of tight lateral retinacular structures or the fascia/muscles of the thigh (predominantly anterolateral). Applied passively by either the patient or clinician, these techniques are designed to optimise movement and facilitate the exercise program. When applied by the clinician, the patient may be in a side-lying position with the knee flexed. The therapist glides the patella medially using the heel of the hand for a sustained stretch (Fig. 36.7). Other simple stretching techniques can be performed by the patient, usually in sitting, with the knee flexed approximately 30–90°. Studies evaluating efficacy of such techniques applied in isolation do not show benefit,¹²¹ supporting the clinical practice to include manual therapy as part of a combined approach to management and not performed in isolation.

Attention must also be paid to improving the compliance of the hip flexors, quadriceps, hamstring and calf muscles as well as the iliotibial band if required, through stretches, soft tissue techniques (Fig. 36.8) or dry needling. Restoration of optimal muscle and fascial length is the goal. Further detail on soft tissue techniques including dry needling are available in Chapter 17.

In-shoe foot orthoses

Prefabricated in-shoe foot orthoses, combined with active retraining of the extrinsic muscles of the foot, improve pain and physical function in people with PFP.¹¹¹ Although the mechanisms underpinning the clinical effects are unclear, it appears both neuromuscular and mechanical factors need to be considered when prescribing in-shoe foot orthoses.¹²² A clinical decision-making paradigm has been proposed to assist with orthosis prescription for people with PFP¹²³ (see also Chapter 8).

Pain reduction during aggravating activities with either anti-pronation taping or an orthosis, may indicate



Figure 36.7 Mobilisation of the patella (supine or side-lying)



Figure 36.8 Soft tissue therapy—myofascial tension of the iliotibial band

which patients are more likely to have a favourable response to orthoses.^{124, 125} Older people with less severe pain and a more mobile midfoot are more likely to experience success with orthosis intervention.¹²³ Additionally, the clinician might consider customised foot orthoses or changing footwear, although the efficacy of these interventions is unknown. These interventions are discussed in Chapter 8.

Braces

Some braces (Fig. 36.9) are commercially available to maintain medial glide. Patella braces reduce patella displacement, increase patella contact area¹²⁶ and reduce PFJ stress¹²⁷ in individuals with PFP. However, an RCT of such a brace did not find any benefit of the brace over a 'sham' knee sleeve or a general quadriceps strengthening program.¹²⁸ Braces are less specific than taping and do not specifically address tilt or rotation; but they may have a role in those patients who are unable to wear tape or who suffer recurrent patella subluxation or dislocation.

Evidence base for physical and exercise interventions

A number of systematic reviews and meta-analyses of randomised clinical trials have assessed the efficacy of physical interventions for PFP (Table 36.8). Level I evidence supports the use of multimodal interventions (combination of retraining, strengthening, balance/coordination, functional retraining, taping and manual therapy)¹¹² and also for individual treatments in isolation.^{112, 129, 130} Proximal (mostly hip) rehabilitation programs, when combined with a quadriceps program, decreases pain in the short and medium term, and emerging evidence suggests that proximal retraining, either combined with quadriceps retraining or in isolation, may be superior to quadriceps retraining alone.¹¹⁵ While tailored patellar taping produces immediate pain reduction, other techniques result in only small (untailored medial patellar taping) or negligible (kinesiotape) pain reductions in the immediate term.

Limited evidence supports better short-term outcomes with foot orthoses versus flat inserts,^{112, 129} versus 'wait and see',¹³¹ and beneficial effects for exercise (versus



Figure 36.9 Patella stabilising brace

control), closed chain exercises (versus open chain exercises) and acupuncture (versus control).¹¹² However, there is moderate evidence for no additive effectiveness of knee braces to exercise therapy on pain and conflicting evidence on function.¹³²

Table 36.8 Best available evidence for the physical and exercise interventions for patellofemoral pain (since 2010)

Intervention	Level of evidence	Reference	Supported by consensus statements ¹¹³ or Best Practice Guide ¹⁰⁸
Combined intervention (exercise, education, manual therapy, taping)	1	Collins 2012 ¹³³ Bolgia 2011 ¹³⁴	Yes ¹¹³ Yes ¹⁰⁸
Exercise therapy	1	van der Heijden 2015 ¹³⁰ Collins 2012 ¹³³ Lack 2015 ¹³⁵ Kooiker 2014 ¹³⁶ Clijisen 2014 ¹³⁷	Yes ¹¹³ Yes ¹⁰⁸
Hip muscle retraining	1	van der Heijden 2015 ¹³⁰ Lack 2015 ¹³⁵ Bolgia 2011 ¹³⁴ Clijisen 2014 ¹³⁷	Yes ¹¹³ Yes ¹⁰⁸

continued

Table 36.8 Cont.

Intervention	Level of evidence	Reference	Supported by consensus statements ¹¹³ or Best Practice Guide ¹⁰⁸
Foot orthoses	1	Barton 2010 ¹²⁹ Bolgia 2011 ¹³⁴ Collins 2012 ¹³³ Hossain 2010 ¹³⁸	Yes ¹¹³ Yes ¹⁰⁸
Patellar taping: immediate effects	1	Barton 2014 ¹¹⁸ Collins 2012 ¹³³	Uncertain ¹¹³ Yes ¹⁰⁸
Kinesiotaping	1	Kalron 2013 ¹³⁹	Not included ¹¹³ Not included ¹¹³
Patellar taping: medium/long-term effect	1	Callaghan 2012 ¹⁴⁰	Uncertain ¹¹³ Yes ¹⁰⁸
Acupuncture	1	Collins 2012 ¹³³	Uncertain ¹¹³ Yes ¹⁰⁸
Gait/movement retraining	5	Barton 2015 ¹⁰⁸	Uncertain ¹¹³ Yes ¹⁰⁸
Mobilisation	1	Collins 2012 ¹³³	No ¹¹³
Bracing	1	Swart 2012 ¹³²	Uncertain ¹¹³ No ¹⁰⁸
Electrotherapeutic agents	1	Lake 2011 ¹⁴¹ Collins 2012 ¹³³	No ¹¹³ No ¹⁰⁸

Blue = supported by meta-analyses (systematic reviews)

Black = expert opinion, clinical guidelines

Orange = insufficient evidence

Red = no evidence supporting its use

Bold text = reviews considered to be moderate or high quality

Although the interventions tested in clinical trials reflect some aspects of clinical practice, in many studies the interventions are not individualised to the patient's needs. Furthermore, some studies have evaluated one aspect of intervention in isolation. These studies have enabled the assessment of a single treatment strategy, but the study results may not be generalised to clinical scenarios. Therefore, while the evidence base for different combinations of physical or exercise therapies for PFP is increasing in quality and quantity, clinicians need to individualise their treatment approaches, thus targeting the treatment to their patient's presentations and requirements and clinician preferences and expertise.¹⁰⁸

Addressing extrinsic contributing factors

Although the patient should reduce the load on the PFJ initially, it is essential that as rehabilitation progresses, any extrinsic factors that may have been placing excessive load on the PFJ (e.g. training, shoes, surfaces) be addressed.

Surgery: to be avoided

Experienced clinicians will have observed that surgery is now much less common for PFP than it was 30 years ago. This is likely to be due to the availability of evidence-based, exercise-based, physical interventions. To date, no surgical RCT has demonstrated the effectiveness of treatments such as chondroplasty or lateral release for PFP. Surgery should only be an option for athletes/patients whom have failed a number of attempts of conservative interventions. Poor surgical outcomes have been reported in the literature.¹⁴²

PRACTICE PEARL

Patient education is an important element of effective treatment. A number of resources exist, including Barton and Rathleff www.patellofemoral.com/completeportscare.com.au/wp-content/uploads/2014/11/Managing-my-patellofemoral-pain_education_single-sheets.pdf.

Patellofemoral osteoarthritis can be managed using an approach similar to patellofemoral pain

Can we prevent patellofemoral OA in people with patellofemoral pain or ACL injury?

Strategies targeting prevention of PFP and traumatic knee injuries are vital to prevent patellofemoral OA in active adults. Following knee injury or PFP, treatments to optimise patellofemoral biomechanics may reduce the risk of patellofemoral OA.

Until more research can determine which features of an individual makes them more likely to develop OA after PFP or knee injuries such as ACL or meniscal, interventions with potential to reduce PFJ stress (e.g. patellofemoral malalignment, low quadriceps strength, higher body mass, valgus knee malalignment) should be included as part of a treatment package. Furthermore, it is probably never too early to discuss the possibility of chronic knee pain, and the need for ongoing management of the knee.

How do we treat people with established radiographic patellofemoral OA?

Once patellofemoral OA has developed, non-surgical management is the preferred primary treatment. Although few clinical trials exist to inform best-practice management, multimodal physiotherapy treatment (i.e. exercise, education, manual therapy and taping)¹⁴³ and knee braces¹⁴⁴ may be efficacious. Patellar bracing may improve PFJ kinematics and knee pain, and shrink BMLs in those with patellofemoral OA.^{145, 146} Although tibiofemoral and patellofemoral OA frequently coexist, it is inappropriate to assume that treatments designed for tibiofemoral OA are optimal for patellofemoral OA.

Can we optimise treatments for patellofemoral OA?

To improve treatments for those with patellofemoral OA, we need to recognise the clinical findings that identify and discriminate them from tibiofemoral OA.²⁴

A number of factors associated with patellofemoral OA may alter the mechanics of the patellofemoral joint and increase joint stress, leading to OA. Identification of these impairments in patients with patellofemoral OA may lead to targeted and personalised interventions, with capacity to not only reduce the symptoms but have potential to slow disease progression. Such factors include:

- **Abnormal patellofemoral joint alignment and trochlear morphology** These are associated with patellofemoral OA (both radiographic and MRI features).
- **Muscle weakness** Quadriceps function, such as muscle size,¹⁴⁷ strength^{148, 149} and muscle force,¹⁵⁰ is impaired in people with patellofemoral OA, and quadriceps weakness is a risk factor for patellofemoral OA.^{151, 152} Impairment in proximal muscle function is also evident on patellofemoral OA.
- **Abnormal biomechanics** Abnormal movement patterns during functional activities may affect PFJ loading and stress,^{153–155} which can lead to OA and pain^{156–158} and are, therefore, potential targets for rehabilitation interventions. There is recent evidence that individuals with patellofemoral OA also demonstrate abnormal biomechanics during gait.^{150, 152, 159–161} In the only longitudinal study to date, Teng et al. found that peak knee flexion moment and flexion moment impulse at baseline lead to progression of PFJ cartilage damage over 2 years.¹⁶²

Note: Patellofemoral pain may predispose to the development of patellofemoral osteoarthritis and hence treatments should aim to not only reduce pain and symptoms but to address potential contributing factors.

How does our treatment need to change for adolescents with patellofemoral pain?

Patellofemoral pain is common in adolescents

Chances are that if you work in clinical setting you will see a lot of young patients with knee complaints. Up to 7% of the adolescent population may be affected (in varying severities) by PFP, making it one of the most common knee conditions among adolescents. Fifty percent of the adolescents with

knee pain will, at some point, seek medical treatment, often through their general practitioner (in a Danish context).¹⁶³ This means that at some point, we are likely to meet them (and their parents) in the clinic. So how should we treat adolescents with PFP? Do they need to be treated differently to adults with PFP?

continued

Education and exercises

Combining exercise therapy and patient education is more effective than patient education alone in adolescents aged 15–19 with PFP.¹⁶⁴ Adding exercise therapy to patient education improves recovery rates in both the short term (3 months) and long term (24 months). However, not all adolescents respond favourably to the exercise therapy, and the overall recovery rate is lower than for adults. This suggests that exercise therapy is effective for some adolescents but not all. How can we improve the effect of exercise therapy among adolescents to ensure good outcomes?

Adherence and modifying the activity level are crucial

Two issues seem to be particularly important when treating adolescents with PFP. One is adherence to exercise therapy, and the second is sports participation.¹⁶⁵ For highly active young adolescents, frequent sports participation is a risk factor for persistent knee pain and failure to modify sports participation may hinder recovery.^{166, 167} This suggests that, in the group of highly active adolescents, PFP may result from excessive loading of the PFJ during sport and physical activity.¹⁶⁵ The clinical implication is that we need to educate both the adolescents and their parents to modify the adolescents' activity levels and make a gradual return to sport that is guided by symptoms.

The second important issue is adherence to exercise therapy. Adolescents with poor adherence (less than 1 exercise session per week) were 4 times less likely to be recovered after 12 months compared to those with good adherence (at least 3 exercise sessions per week).¹⁶⁴

To improve adherence, it is essential that the adolescent and the parent are educated about the importance of adherence and understand why the exercises are important. If you suspect that the adolescent may not perform the prescribed home exercises, it might be relevant to focus on supervised exercises to improve outcomes.

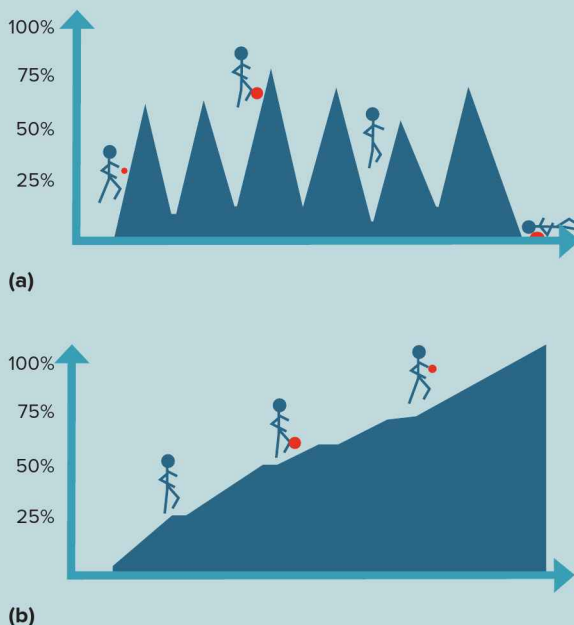


Figure 36.10 a and b Modifying the physical activity level may be key to successful treatment and to avoid recurrence of knee pain.

PATELLOFEMORAL INSTABILITY

Before discussing the causes of and treatments for, patellofemoral instability we need to clearly define the term. As is also the case at shoulder, 'instability', is not a synonym for 'dislocation'. Instability refers to excessive joint range of motion (and is often referred to as 'subluxation'). Patellofemoral instability may be primary (referred to as patellar subluxation) or secondary (meaning that the instability is due to previous acute patella dislocation) (see Chapter 35).

Primary patellofemoral instability

Factors that predispose to this condition include patella alta (a patella that is located more superiorly than normal), trochlear dysplasia and generalised ligamentous laxity. Patients usually describe sensations of instability (patella moving or slipping) and the pattern of tenderness around the patella may be similar to PFP or more generally

distributed. Examination reveals patella hypermobility with apprehension and pain when the patella is pushed laterally by the clinician.

Treatment

Treatment of patellofemoral instability parallels that of PFP. The aim of acute management is to reduce pain and swelling. Patellar taping in combination with a knee extension brace may provide temporary immobilisation after an initial episode. The patient may use crutches for either partial or

PRACTICE PEARL

Rehabilitation to improve active support for the PFJ is paramount as dominant predisposing factors cannot be addressed easily (bony morphology and ligamentous laxity).

non weight-bearing in that instance. Rehabilitation requires a comprehensive multimodal approach, as outlined earlier for PFP to be commenced as soon as possible.

In acute situations, biofeedback and/or electrical stimulation may be required as an adjunct to vasti retraining. Taping (and/or bracing) may be part of long-term management. It is also vital to address any proximal and distal contributing factors (see above), maximise trunk and lower limb muscle strength and/or endurance and provide advice regarding avoiding aggravating activities.

Surgery is rarely indicated. Surgical approaches aim to correct the structural/anatomical predisposing factors, and techniques used include arthroscopic medial plication. Bony realignment procedures including 'trochleaplasties' have largely lost popularity in Scandinavia and Australia, but are performed in a limited number of countries including the United States. These interventions often do not have favourable outcomes, thus they should only be considered if a properly managed non-surgical program fails. An intensive rehabilitation program should be trialled prior to the athlete considering surgical options.

Secondary patellofemoral instability

Secondary instability results from a primary dislocation episode (see Chapter 35 for acute management) that is likely to have arisen because of rupture of the medial patellofemoral ligament. This ligament is the main static restraint to lateral patella translation (Fig. 35.17). It acts like a guy rope on a tent. Individuals with persistent patellofemoral instability after an acute dislocation may require additional investigations. Imaging (Fig. 35.18) may reveal evidence of osteochondral damage to the articular surface of the patella and femur as well as predisposing anatomical abnormalities.

Arthroscopy may be required to remove a loose osteochondral fragment, and, if appropriate, the medial patellofemoral ligament may need reconstruction.¹⁶⁸ As with surgery for primary instability, surgery for secondary instability also requires aggressive rehabilitation. Rehabilitation for secondary patellofemoral instability follows the same guidelines as primary patellofemoral instability. At the current time, there is limited evidence to suggest that surgical management of primary patella dislocation decreases the risk of recurrent dislocation.¹⁶⁹

PATELLAR TENDINOPATHY

There have been substantial advances in understanding the management of this condition in the past decade. Successful management of the athlete with patellar tendinopathy requires thorough clinical and functional assessment and then appropriate load management.

Nomenclature

Patellar tendinopathy is otherwise known as 'jumper's knee' due to its frequency in jumping sports (e.g. basketball, volleyball and high, long and triple jumps).¹⁷⁰ However, the condition also occurs in athletes who change direction, as well as those who put large energy storage loads on their tendon, such as running downhill. The term 'patellar tendinitis' is a misnomer as the pathology underlying this condition is degenerative tendinosis rather than a purely inflammatory condition (see Chapter 4).¹⁷¹ On balance, patellar tendinopathy is probably the most appropriate general label for this condition as it describes the clinical condition of pain and loss of function, independent of the underlying pathology.¹⁷²

Clinical features

The clinical features of patellar tendinopathy are outlined in Table 36.2. The key to making an accurate clinical diagnosis is that the patient complains of well-localised (focal) pain at the attachment of the patellar tendon to the patella (Fig. 36.11). The tendon is frequently thickened. The pain is aggravated by jumping, changing direction and decelerating. Distal lesions are less common and midsubstance lesions are rarely reported.¹⁷³

Clinicians should assess factors such as weakness of the lower limb musculature, including calf, quadriceps and gluteal muscles. This loss in muscle strength often results in dysfunction of the kinetic chain, especially in its ability to function as a spring. These deficits may predispose or be a consequence of pain, but either way are important to assess so that they may be addressed in rehabilitation.

PRACTICE PEARL

Pain on palpation (i.e. tenderness) is not a useful test to diagnose tendinopathy in general (Chapter 14); it is not a helpful diagnostic test in patellar tendinopathy as pain at this site is common and does not indicate the pain is from the tendon. Most importantly, it is not a reliable outcome measure to assess response to rehabilitation.

Note that the proximal patellar tendon is often tender in jumping athletes with normal tendons therefore palpation has poor specificity for patellar tendinopathy. Also, some athletes with patellar tendon pathology are not tender at the junction of patellar tendon and the patella—hence the test does not have perfect sensitivity. Note that palpation is also not a useful test of recovery (outcome measure)—this is discussed later.

To make the diagnosis of patellar tendinopathy, the clinician must be able to reproduce the patient's focal

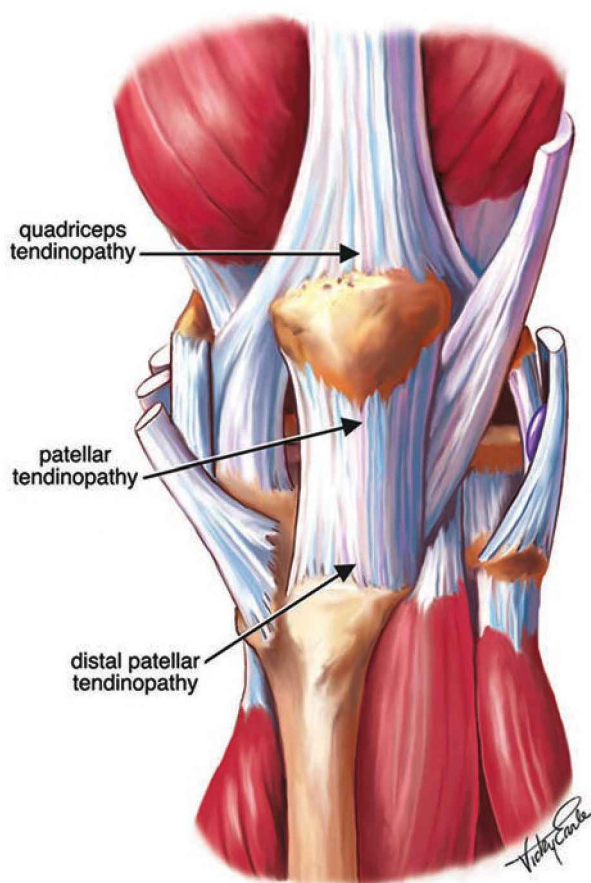


Figure 36.11 Sites of tendon pathology in the knee extensor mechanism. The most common presentation is patellar tendonopathy. Quadriceps tendonopathy is discussed later. Pain at the distal tendon (tibial tuberosity) can be associated with Osgood–Schlatter disease (Chapter 44)

pain when placing high loads on the tendon. The clinician should progressively increase the tendon load with functional activities, such as a squat, then progressing to a hop, to test the patient's ability to complete the task well and to reproduce the pain. In patellar tendonopathy the pain remains localised. If the pain becomes diffuse with loading, consider other diagnoses.

PRACTICE PEARL

Knee pain that is described as being diffuse with loading does not represent a clinical diagnosis of patellar tendonopathy (or tendon pain). Patients can perceive pain from the PFJ in the region of the patellar tendon.

In patellar tendonopathy, a useful outcome measure (Chapter 16) is pain on the decline squat (Fig. 36.2d). Document the amount of pain and the range when it occurs. An additional method to monitor the clinical progress of patellar tendonopathy is the VISA-P questionnaire (Table 36.9).^{174, 175} This simple patient-completed questionnaire takes less than 5 minutes to complete.

Investigations

Ultrasound and MRI can image patellar tendon pathology. These imaging modalities have low sensitivity with better specificity for patellar tendonopathy (Fig. 36.12). Imaging findings need to be placed within the context of the clinical presentation and the loading history of the patient. Up to 50% of asymptomatic players in patellar tendon loading sports (e.g. volleyball, basketball) have imaging abnormalities.²² These changes are best described as load-appropriate changes and do not contribute to making a clinical diagnosis.

Management: is the athlete still competing?

After making the diagnosis of patellar tendonopathy, the approach to management will vary if the player can keep competing or if the tendonopathy is at a level where the player needs to withdraw from competition and complete rehabilitation.

Factors that determine whether a player should keep playing or withdraw from competition are: the levels of (i) pain, (ii) function and (iii) effect on performance (Table 36.10). Players always want to keep playing and the clinician will support this where appropriate, but there are also times when the quality shared decision is for the player to withdraw from training and competition and undergo comprehensive rehabilitation.

As the player will want to keep competing in most cases, we address that scenario first. We discuss out-of-competition rehabilitation below.

The approach to 'in-season' treatment of the player with patellar tendonopathy

When the player presents with pain in-season, the clinician should discuss the key elements of treatment and explain that there is unlikely to be complete relief of pain during the season.

In-season management of patellar tendonopathy requires maintenance (or improvement) of strength in the affected muscle-tendon unit and the rest of the kinetic chain. Load reduction, correcting biomechanical factors and adjunct therapies such as soft tissue therapy to the affected muscle (not the tendon) are also appropriate in this scenario (detailed below).

Table 36.9 Victorian Institute of Sport Assessment (VISA-P) questionnaire (English version, this is available in numerous languages)¹⁵⁰

1. For how many minutes can you sit pain-free? POINTS

0 min												100 min	
	0	1	2	3	4	5	6	7	8	9	10		

2. Do you have pain walking downstairs with a normal gait cycle? POINTS

Strong severe pain												No pain	
	0	1	2	3	4	5	6	7	8	9	10		

3. Do you have pain at the knee with full active non-weight-bearing knee extension? POINTS

Strong severe pain												No pain	
	0	1	2	3	4	5	6	7	8	9	10		

4. Do you have pain when doing a full weight-bearing lunge? POINTS

Strong severe pain												No pain	
	0	1	2	3	4	5	6	7	8	9	10		

5. Do you have problems squatting? POINTS

Unable												No problems	
	0	1	2	3	4	5	6	7	8	9	10		

6. Do you have pain during or immediately after doing 10 single-leg hops? POINTS

Strong severe pain												No pain	
	0	1	2	3	4	5	6	7	8	9	10		

7. Are you currently undertaking sport or other physical activity? POINTS

0 ☐ Not at all

4 ☐ Modified training ± modified competition

7 ☐ Full training ± competition but not at same level as when symptoms began

10 ☐ Competing at the same or higher level as when symptoms began

8. Please complete EITHER A, B, or C in this question.

- If you have no pain while undertaking sport please complete Q8A only.
- If you have pain while undertaking sport but it does not stop you from completing the activity, complete Q8B only.
- If you have pain that stops you from completing sporting activities, please complete Q8C only.

A. If you have no pain while undertaking sport, for how long can you train/practice? POINTS

	Nil		1–5 min		6–10 min		11–15 min		>15 min	
	0		7		14		21		30	

OR

B. If you have some pain while undertaking sport, but it does not stop you from completing your training/practice, for how long can you train/practice? POINTS

	Nil		1–5 min		6–10 min		11–15 min		>15 min	
	0		4		10		14		20	

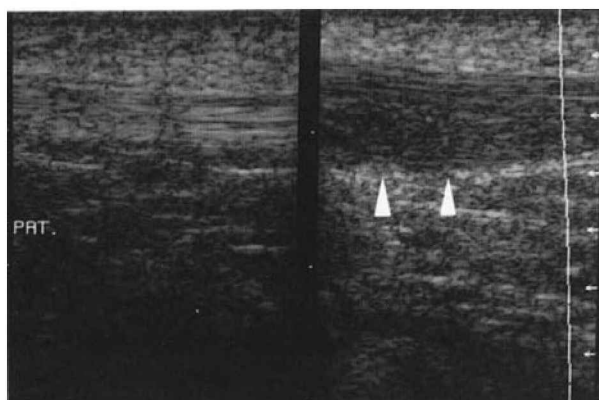
OR

C. If you have pain that stops you from completing your training/practice, for how long can you train/practice? POINTS

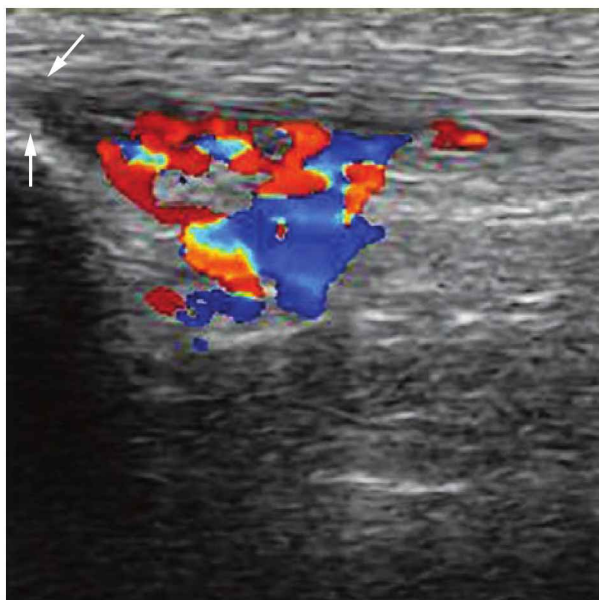
	Nil		1–5 min		6–10 min		11–15 min		>15 min	
	0		2		5		7		10	

TOTAL SCORE

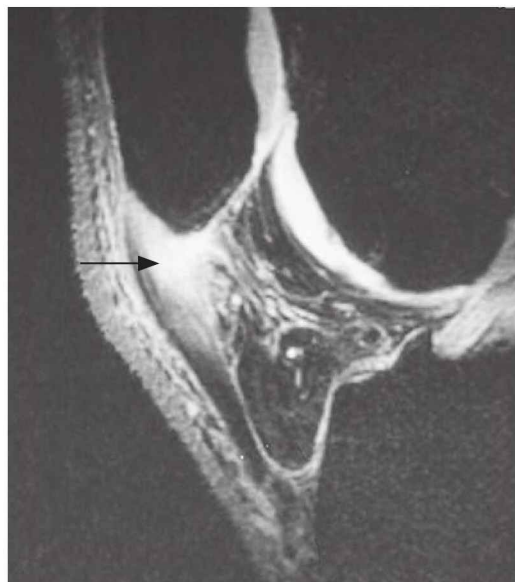
/100



(a)



(b)



(c)

Figure 36.12 Ultrasound and MRI images of patellar tendinopathy in athletes (a) Ultrasound image (normal [left] and thickened [right] tendon). Arrowheads point to the posterior/deep edge of the patellar tendon (b) Ultrasound image with colour Doppler showing abnormal vascularity (blue and red signal) near the junction of the patellar tendon and the patella (arrows). (c) Sagittal slice MRI appearance of patellar tendinopathy

In-season treatment *does not include* heavy eccentric loading, power, and energy storage and release exercises, as the tendon is subjected to these frequently during training and competition. Injection therapies are contraindicated in-season because they require around 2–3 weeks out of sport after injection.

Table 36.10 Factors to take into account when making the decision about whether a player should keep competing or stop playing because of patellar tendinopathy. The decision should be shared between player and clinician

	Pain?	How limited is function?	Player's performance level	Likely clinical decision	Management strategy
Scenario 1 (often long-term)	Severe	Severely limited	Poor	Unable to compete—player understands he/she cannot play	Begin comprehensive out-of-competition rehabilitation program
Scenario 2 (often medium-term)	Tolerable	Moderate to severely limited	Adequate	Continue to compete but player is at risk of aggravating the tendon or suffering other injury	Maintain or improve function (see below)
Scenario 3 (more recent onset of symptoms)	Severe	Moderately limited	Adequate	Continue to compete but modify training/games to lower load	Use pain-relieving strategies (isometric exercise, medication)

This section outlines the physiotherapy approach to in-season management.

Relative load reduction: modified activity and biomechanical correction

There are numerous ways of reducing the load on the patellar tendon. The player must modify his or her training and perhaps game time, with reduction in the amount of jumping or sprinting in a session, the number of high load days each week or the total weekly training hours. Absolute rest is contraindicated and some continued lower limb load that does not include excessive energy storage loads (i.e. jumping, changes of direction, deceleration) on the tendon is critical to maintain tendon integrity, kinetic chain function and cardiovascular fitness.

Optimising biomechanics to improve the energy-absorbing capacity of the limb should be directed at both the affected musculotendinous unit and the hip and ankle. The ankle and calf are critical in absorbing the initial landing load and reduces the load transmitted to the knee.¹⁷⁶ Biomechanical studies reveal that only about 40% of landing energy is transmitted proximally.¹⁷⁷ Thus, the calf complex, and especially the soleus, must function well to prevent more load than necessary being placed on the patellar tendon.

Biomechanical correction should include assessment of both anatomy and function. Better landing techniques can decrease patellar tendon load. Compared with flat-foot landing, forefoot landing generates lower ground reaction forces and, if this technique is combined with a large range of hip or knee flexion, vertical ground reaction forces can be reduced by a further 25%.¹⁷⁷ Landing with weight further forward that uses all available dorsiflexion may also decrease patellar tendon load.^{178, 179} Anatomical variants that may contribute to persistent patellar tendinopathy include limited ankle dorsiflexion and hallux rigidus. Tight hamstrings, quadriceps and calf muscles may increase the load on the patellar tendon. However, solely focusing on biomechanical correction without addressing kinetic chain strength and capacity will ultimately deliver an undesired outcome.

Isometrics for pain relief

Isometric exercises reduce pain and improve motor drive by reducing cortical inhibition.⁴ (See also Chapter 4, Fig. 4.14) These exercises are best performed on weighted machines (Fig. 36.13) with single-leg isolated knee joint movement. Isometric exercise also is as effective as isotonic exercise in reducing pain in-season.^{180, 181}

The specific parameters for isometric exercise are 45-second holds at 70% maximal voluntary isometric quadriceps contraction (MVC) with 2 minutes between each of the 5 repetitions. Long holds at mid-range with heavy weights are essential to the protocol but individual variations may be required.



Figure 36.13 Isometric holds on a knee extension machine. Hold the highest weight possible for up to 45 seconds with good control

Strengthening

Strategies that maintain or improve strength of the entire kinetic chain are critical when managing a player during the season. Slow heavy-resistance training to maximise muscle strength prevents unloading weakness and atrophy, but may not improve performance or pain when competing.^{182, 183}

Particular attention can be paid to the calf and gluteal muscles as these attenuate landing forces as well as improving quadriceps strength on the unaffected side. Cross-education means that quadriceps strength on the unaffected side can improve the strength of affected quadriceps by up to 20%.

The effect on cortical inhibition suggests that isometric exercises should also be used prior to quadriceps strength training throughout the four-stage program (Table 36.11) to improve the motor drive to the quadriceps.

Decline squat exercises (Fig. 36.14a) have traditionally been widely used, but are painful and can be detrimental when used during the season.^{184, 185}

Soft tissue therapy

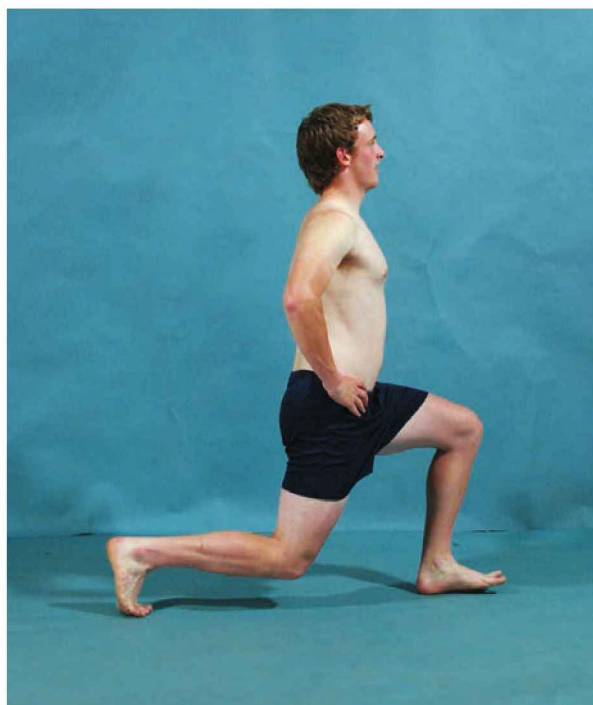
The use of friction to treat patellar tendinopathy remains a popular treatment, but lacks a logical theoretical construct and can further irritate the tendon. Studies that compared soft tissue therapy/transverse friction to other treatments demonstrated little benefit in reducing pain.^{186, 187} Digital ischaemic pressure and sustained myofascial tension to tight muscles or trigger points in the quadriceps, hamstrings and calf muscles can be performed.

Pain reduction

Several strategies exist for reducing pain aside from the isometrics discussed earlier, all have to be considered against their potential to further irritate the tendon.

Table 36.11 Four-stage strengthening program for treatment of patellar tendinopathy

Stage	Type of overload	Timing	Purpose	Activity	Maintain/kinetic chain exercise
1	Heavy load isometric exercise	0–1 months	Reduce pain	Weight-machine-based isolated quadriceps exercise	Begin isotonic exercise for calf and gluteals as well as quadriceps exercise for the other leg
2	Heavy load isotonic exercise	1–3 months	Improve strength, regain muscle bulk	Weight-machine-based heavy isotonic exercise	Isometrics before strength to reduce cortical inhibition Kinetic chain exercises
3	Energy storage exercises	3–4 months	Progressive high tendon load	Fast, body-weight-only knee flexion exercise to load the patellar tendon spring	Maintain isotonic exercise
4	Energy storage and release exercises	4–6 months	Sport-specific high tendon load	Jumping and change-of-direction exercises	Stop stage 3 exercise, maintain heavy load isotonic exercise
Return to training and competition	Sport-specific training	6+ months	Controlled-load return to sports activity	Sports drills with attention to frequency (2–3 times a week), volume and intensity	Maintain isotonic exercise

**(a)****(b)****Figure 36.14** Patellar tendon eccentric strength training exercises (a) Single-leg squat on decline board (may aggravate pain and thus be detrimental in-season) (b) Lunge



(c)

Figure 36.14 (cont.) (c) Lunge with weights**Cryotherapy**

Cryotherapy (e.g. ice) is a popular adjunct to treatment but if the athlete finds no clinical benefit from this modality, there is no rationale for persisting.

Pharmacotherapy

Studies of pharmacotherapy in the treatment of patellar tendinopathy are limited to phoresis as few studies have investigated oral medications. The use of aprotinin in tendinopathy has some evidence, but is not recommended due to the risk of anaphylaxis after injection. Iontophoresis with corticosteroid improved outcome compared to phonophoresis, suggesting it may introduce corticosteroid into target tissue more effectively than phonophoresis.¹⁸⁸

The model for tendinopathy proposed by Cook and Purdam suggests that early stage (reactive) tendinopathy could respond to medications that reduce cell activity and protein production.¹⁸⁹ Ibuprofen, doxycycline and green tea are hypothesised to improve pain and pathology in tendinopathy.¹⁹⁰

Taping

Inferior patellar tendon taping reduced pain minimally and no more than placebo taping.¹⁹¹

Extracorporeal shock wave therapy

Extracorporeal shock wave therapy has been used to treat tendon pain. A systematic review including two RCTs

showed variable results of randomised controlled studies; one RCT showed no benefit and the other showed better outcomes versus placebo in long-term tendon pain.¹⁹²

Extracorporeal shock wave therapy may cause a neuropraxia to the nociceptive nerves resulting in decreased pain for several weeks. There is some animal evidence that it can have a negative effect on normal tendon structure within hours that remains for several weeks.¹⁹³

The approach to out-of-competition treatment of the player with patellar tendinopathy

If the player needs to stop competing (see Table 36.10) an extensive out-of-competition exercise program is required to restore tendon, muscle and kinetic chain function. Resolution of the pathology is not required for recovery and good clinical outcomes can arise with limited changes in tendon structure.¹⁹⁴⁻¹⁹⁶

Another important concept that supports load-based rehabilitation is that a pathological tendon has adapted and has more aligned tendon structure than a structurally normal tendon.¹⁹⁷ This suggests that directing treatments at the area of pathology may be unnecessary and positive clinical outcomes may be due to improving the capacity of already present normal tendon structure.

The clinician will discuss the key elements of management and time frames for recovery/eventual return to play with the athlete, emphasising that there are no short cuts to recovery of function.

Conservative quadriceps rehabilitation of patellar tendinopathy requires appropriate strengthening, power, energy storage and release exercises, load reduction, and correction of biomechanical errors, along with adjunct therapies such as soft tissue therapy to the affected muscle (not the tendon). More invasive treatments, such as injection with substances that either affect the vessels (sclerotherapy with polidocanol), theoretically promote tendon repair (platelet rich plasma [PRP] injections) or augment matrix structure (prolotherapy), lack evidence.¹⁹⁸⁻²⁰⁰

Relative load reduction

There are numerous ways of reducing the load on the patellar tendon without resorting to complete rest or immobilisation. Absolute rest is contraindicated; early rehabilitation with isometric and resistance training not only recovers muscle strength but reduces pain, so can be used immediately in the rehabilitation process. Variations on the isometric protocol outlined above may be required for individuals with high levels of pain. Although it is clear that long holds with heavy weights are required, it is unknown what is the most important (i.e. length of hold or external weight).

When and how should patellar tendon strengthening begin?

Therapists often have concerns as to when and how they should begin a strengthening program. Even athletes with the most severe cases of patellar tendinopathy should be able to begin some weight-based strength and other exercises (such as calf strength and isometric quadriceps work) in standing. However, the athlete who has not lost a lot of knee strength and bulk can progress quickly to the speed part of the program.

Both pain and the ability of the musculotendinous unit to do the work should guide the amount of strengthening to be done. If pain is a limiting factor, then the program must be modified so that the majority of the work occurs without aggravating symptoms within 24 hours of the exercise. A subjective clinical rating system, such as the VISA questionnaire (Table 36.9), administered at about monthly intervals, will help both the therapist and the patient measure progress.

If pain is under control, it is essential to monitor the ability of the limb to complete the exercises with control and quality. Exercises should only be progressed if the previous work load is easily managed, pain is controlled, and function is satisfactory.

Athletes with patellar tendinopathy tend to 'unload' the affected limb to avoid pain, so they commonly have not only weakness but also abnormal motor patterns that must be reversed. Strength training must graduate quickly to incorporate single-leg exercises (Fig. 36.14) as the athlete

can continue to unload the affected tendon when exercising using both legs. Thus, exercises that target the quadriceps specifically (such as single-leg extensions) may have a place in the rehabilitation of patellar tendinopathy. Similarly, when the athlete is ready, increase the load on the quadriceps by having the patient stand on a 25° decline board to do squats. Compared with squatting on a flat surface, this reduces the calf contribution during the squat.

The therapist should progress the regimen by adding load and speed and then endurance to each of those levels of exercise. Combinations such as load and speed, or height and load, then follow. These end-stage exercises can provoke tendon pain, and are only recommended after a prolonged rehabilitation period, and when the sport demands intense loading. In several sports, it may not be necessary to add potentially aggravating activities such as jump training to the rehabilitation program, whereas in volleyball, for example, it is vital.

Finally, the overall exercise program must correct aberrant motor patterns such as stiff landing mechanics (discussed above) and pelvic instability. For example, weight-bearing exercises must be in a functionally required range, and the pelvis position must be monitored and controlled at all times. The common errors in rehabilitation strength programs are listed in Table 36.12.

Table 36.12 Why rehabilitation programs fail at various stages

Early failure	Late failure
Insufficient strength training	Failure to monitor the patient's symptoms
Progression of rehabilitation program too rapid	Rehabilitation and strength training end on return to training, instead of continuing throughout the return to sport
Inappropriate loads during rehabilitation (too little, too much)	No speed rehabilitation
	Plyometrics training performed inappropriately, not tolerated, or unnecessary

Restoring function

Strength, power and energy storage and release capacity are required for all the muscles, especially the anti-gravity muscles—the calf, quadriceps and gluteal muscles. An effective rehabilitation program embraces the principles outlined in Table 36.11. Commonly prescribed exercises are illustrated in Figure 36.14. When and how a strengthening program should begin is outlined in the box.

Several papers have documented the effectiveness of quadriceps strengthening exercises on patellar tendinopathy.^{8, 201–205} These can be divided into two groups: those studies that investigated eccentric exercises on a decline board and those using other exercises.

The studies that have investigated the effectiveness of exercise on a 25° decline board—a method specifically loading the extensor mechanism of the knee by increasing the moment arm of the knee²⁰³—include two randomised trials which reported improvements in pain, function and return to sport with exercise, although time frames for improvement varied.^{8, 204} In another study, compared to surgery, eccentric exercise on a decline board provided similar outcomes at 12 months.²⁰⁵ Care with these exercises is required in the early painful stages as they can be provocative.

Other studies have investigated heavy load strengthening and report equally good results.²⁰⁶ Three papers

have suggested that exercise-based interventions such as squatting, isokinetics and weights reduced the pain of patellar tendinopathy.^{186, 201, 202}

Soft tissue therapy, cryotherapy, pharmacotherapy, taping

Massage can be used for muscle soreness resulting from resistance training. Ice is of little benefit during rehabilitation, and medications and taping also have no place in this phase.

Biomechanical correction

Ongoing management of any range of movement deficits in the ankle and foot can be continued early in the rehabilitation program. Formal correction of running and jumping techniques, if required, are commenced once kinetic chain function has been mostly restored. However, solely focusing on biomechanical correction without addressing kinetic chain strength and capacity will ultimately deliver an undesired outcome.

Injection therapy

Neovascularisation is a cornerstone of degenerative tendon pathology and is the target of treatment by Alfredson and Ohberg's sclerosing injection.²⁰⁷ In a high-quality RCT of elite athletes with patellar tendinopathy, investigators found that sclerosing injections with polidocanol resulted in a significant improvement in knee function and reduced pain in patients with patellar tendinopathy.²⁰⁸ However, polidocanol has a neurotoxic effect that may underpin its effectiveness, rather than altering the vascularity.²⁰⁹ This is supported by research that shows tendons that have substantial vascularity can remain pain-free.

Corticosteroid injections (CSI) have been compared to placebo and CSI improved symptoms so much so that all 12 athletes who had placebo injection subsequently had CSI. It is worthwhile noting that after initial improvement, 50% of athletes in the CSI group failed to return to sport and were referred to surgery.²¹⁰ Blood injection therapy (autologous blood, platelet rich plasma [PRP]) is being used clinically, but controlled trials show no additional benefit over placebo injection.²¹¹ High volume injections and the addition of tendon cells are being used clinically with little or no quality research to show their efficacy.

Surgery

The only randomised trial to date that compared surgical treatment and conservative management of patellar tendinopathy was published in 2006.²¹² There was no significant difference in outcome between groups; thus, surgical intervention provided no benefit over conservative

management. Thus, the clinical implication is that surgery is not a 'quick fix' for patellar tendinopathy.²¹²

A number of different surgical techniques have been proposed. There has been some enthusiasm for arthroscopic debridement of the anterior portion of the fat pad adjacent to the patellar tendon and results published to date appear similar to those of patients undergoing open surgery.^{213, 214} Open surgery aims to remove the region of tendinosis from the proximal patellar tendon. A systematic review of 17 studies of surgery for patellar tendinopathy reported that outcomes were reasonable after both open and arthroscopic surgery, but that return to sport was faster with the arthroscopic surgery.²¹⁵

We recommend surgery only after a thorough, high-quality rehabilitation program has failed. Surgeons must advise patients that while symptomatic benefit is very likely, return to sport at the previous level cannot be guaranteed (60-80% likelihood).^{213, 216} Time to return to the previous level of sport, if achieved, is likely to take between 6 and 12 months.^{213, 216}

Sudden, acute patellar tendon pain

The term 'partial tear' is often used when the player has a sudden painful episode, which may be associated with disability. A small partial tear of the patellar tendon is often diagnosed by ultrasonography following an event like this or as an incidental finding on ultrasound examination, and it is difficult to differentiate a tear from an area of tendinosis.^{217, 218} Based on surgical specimens in the Achilles tendon, the ability of ultrasound to differentiate degenerative changes and partial tears are limited.

This type of partial tear occurs in degenerative tendinopathy and can be managed conservatively. There are few consequences in the patellar tendon as the tissue trauma is likely to be in an area of pathology of the patellar tendon and not in the normal, load-bearing tissue. If the pain and dysfunction after acute onset pain persists, further imaging and possible intervention may be justified to stimulate some healing response in the tendon.

LESS COMMON CAUSES OF ANTERIOR KNEE PAIN

Fat pad irritation/impingement

Fat pad syndrome was first described by Hoffa in 1903 to describe a condition where the infrapatellar fat pad was impinged between the patella and the femoral condyle due to a direct blow to the knee. More commonly, fat pad irritation occurs with repeated or uncontrolled hyperextension of the knee where it is impinged between the femur and tibia. The condition can be extremely painful and debilitating, as the fat pad is one of the most pain-sensitive structures in the knee.²⁵

Chronic fat pad irritation is relatively common and often goes unrecognised. The pain is often exacerbated by extension manoeuvres, such as straight leg raises and prolonged standing, so it needs to be recognised early so that appropriate management can be implemented.

Clinical findings include localised tenderness (at the inferior pole of the patella, deep to the tendon) and puffiness in the fat pad with the inferior pole of the patella appearing to be (or actually being) displaced posteriorly. Pain may be reproduced with active knee extension and passive overpressure in extension. Contracting the quadriceps with the knee extended may aggravate the pain during the acute phase. There may be quadriceps weakness. Patients often have hyperextension of the knees (*genu recurvatum*) associated with increased anterior pelvic tilt.

Thus, treatments should be directed to improve the control, strength and endurance of local (*vasti*), proximal and distal muscles as well as retraining to avoid uncontrolled and excessive terminal knee extension manoeuvres.

A popular clinical approach consists of treating the inferiorly tilted patella by taping across the superior surface of the patella to lever the inferior pole forward and relieve impingement of the fat pad (Fig. 36.6d). Unloading of the fat pad may be required to relieve the symptoms further.

To unload the fat pad, a 'V' tape is placed below the fat pad, with the point of the 'V' at the tibial tubercle coming wide to the medial and lateral joint lines. As the tape is being pulled towards the joint line, the skin is lifted towards the patella, thus shortening the fat pad (Fig. 36.15). Improving lower limb biomechanics is often helpful. Surgery should be avoided if possible. To date, there have been no RCTs of surgery for this condition.²¹⁹

Osgood–Schlatter lesion

Osgood–Schlatter lesion is an osteochondrosis (Chapter 44) that occurs at the tibial tuberosity in children as they progress through puberty. It is much more common among boys of about 13–15 years (but these ages vary) than girls (age about 10–12) and results from excessive traction on the soft apophysis of the tibial tuberosity by the powerful patellar tendon. It occurs in association with high levels of activity during a period of rapid growth and is associated with a change in the tendon. Longitudinal imaging of the tendon in adolescents without pain shows that the attachment transitions to a normal attachment through a process that can be interpreted as osteochondritic.²²⁰ As with all tendon-related pain, the clinician must be careful not to diagnose based on imaging findings.

Treatment consists of reassurance that the condition is self-limiting and muscle strength exercises. Whether or not to play sport depends on the severity of symptoms. Children with mild symptoms may wish to continue

to play some or all sport; others may choose some modification of their programs. If the child prefers to cease sport because of pain, that decision should be supported. However, the amount of sport played does not seem to affect the time taken for the pain to disappear.

Sinding–Larsen–Johansson lesion

This rare lesion is one of the group of osteochondroses found in adolescents (Chapter 44). It is a clinically unimportant differential diagnosis in young patients with pain at the inferior pole of the patella, who can actually have patellar tendinopathy at a young age.

Quadriceps tendinopathy

Pain arising at the quadriceps tendon at its attachment to the patella occurs mainly in the older athlete and in weightlifters as the quadriceps tendon is loaded more in a deeper squat. It is characterised by tenderness at the superior margin of the patella and pain on resisted quadriceps contraction. Treatment follows the same principles as treatment of patellar tendinopathy. Differential diagnosis is suprapatellar pain of PFJ origin, plica and bipartite patella. Ruptures of the entire quadriceps attachment to the patella are not uncommon in middle-aged men and require surgery and extensive rehabilitation.

Bursitis

There are a number of bursae around the knee joint (Fig. 36.16). The most commonly affected bursa is the pre-patellar bursa. Pre-patellar bursitis ('housemaid's knee') presents as a superficial swelling on the anterior aspect of the knee. This must be differentiated from an effusion of the



Figure 36.15 Fat pad unloading tape. Tape is applied in a 'V' from the tibial tuberosity to the joint lines. The fat pad region is pinched to unload the fat pad while applying the tape. This tape is often combined with taping of the superior pole of the patella (Fig. 36.6d) in the treatment of fat pad impingement

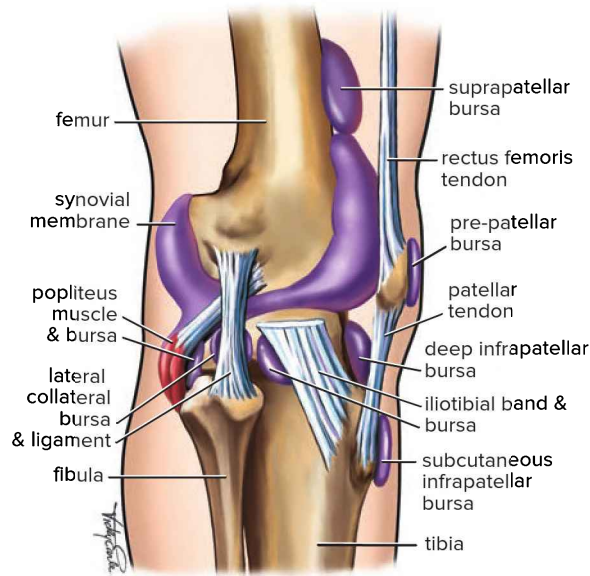


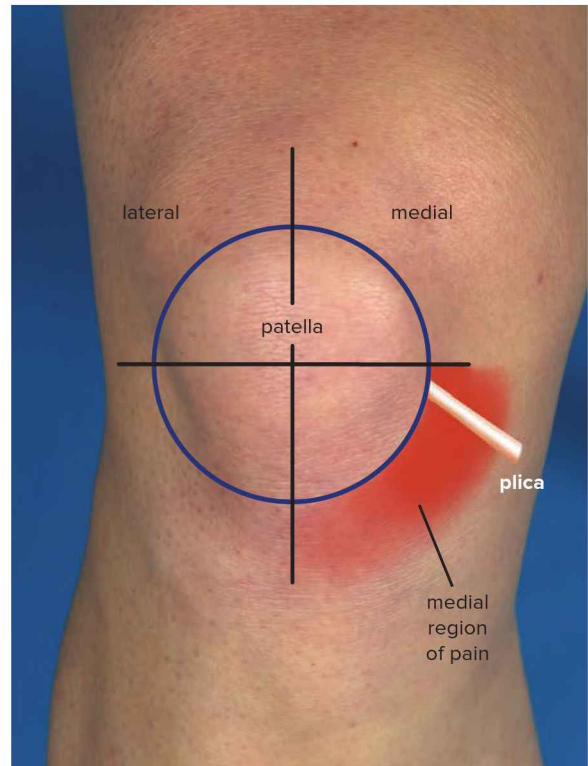
Figure 36.16 Bursae around the knee joint

knee joint. Acute infective pre-patellar bursitis, common in those who kneel a lot, should be identified and treated quickly. Infrapatellar bursitis can also cause anterior knee pain that may mimic patellar tendinopathy; this bursa forms part of an enthesis organ of the distal insertion and, thus, can be challenging to treat. Pes anserinus bursitis is often associated with an underlying rheumatological condition.

Treatment of mild cases of bursitis includes non-steroidal anti-inflammatory drugs (NSAIDs). More severe cases require aspiration and infiltration with a corticosteroid agent and local anaesthesia, followed by appropriate treatment of the enthesis if appropriate.

Synovial plica

The importance of the synovial plica, a synovial fold found along the supero-medial edge of the patella (see Fig. 36.17), has been a matter of considerable debate. An inflamed plica may cause variable sharp pain located anteriorly, medially or posteriorly. The patient may complain of sharp pain on squatting and it can mimic quadriceps tendinopathy. On examination, the plica is sometimes palpable as a thickened band under the medial border of the patella. It should only be considered as the primary cause of the patient's symptoms when the patient fails to respond to appropriate management of PFP. In this case and in the presence of a tender thickened band, arthroscopy should be performed and the synovial plica removed.



(a)



(b)

Figure 36.17 Synovial plica (a) Medial synovial plica presents with pain in the region highlighted (red). A band of tissue may be palpable (+/- tender) running from the medial patella to the epicondylar region (blue) (b) A frayed, thickened synovial plica photographed at arthroscopy. It catches between the patella above it and the femoral condyle

REFERENCES

References for this chapter can be found at www.mhhe.com/au/CSM5e